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SURVEY

Wireless Sensor Networks: From Fundamentals and Applications to Innovations and Future Trends

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ABSTRACT Wireless Sensor Networks (WSNs) have transformed data acquisition and processing across diverse domains such as environmental monitoring, healthcare, and industrial automation. Recent advancements in areas such as federated intelligence, blockchain-based trust mechanisms, and quantum sensing are reshaping the landscape of WSNs. Yet, their full implications for next-generation applications demand deeper exploration. This survey is driven by the need to synthesize these emerging directions through a forward-looking and interdisciplinary lens, highlighting their transformative potential in evolving WSNs into intelligent, autonomous systems. It examines key solutions in energy harvesting, adaptive communication, and edge intelligence, which address persistent constraints like limited energy availability, dynamic topologies, and scalability. Moreover, it highlights the integration of WSNs with disruptive technologies such as the Internet of Things (IoT), artificial intelligence (AI), and digital twins, enabling intelligent, autonomous systems capable of real-time analytics. Critical energy optimization, security, and data management challenges are explored alongside cutting-edge solutions like federated learning (FL), quantum sensing, and blockchain-based trust mechanisms. Future research directions emphasize bio-inspired architectures, hybrid communication in extreme environments, and efficient multi-modal data fusion. This survey serves as a foundational reference for researchers and practitioners seeking to design resilient, intelligent WSN-based systems for the next generation of applications.

INDEX TERMS Wireless sensor networks, adaptive communication protocols, Internet of Things, smart systems, data processing.

I. INTRODUCTION

WSNs have emerged as a transformative technology for distributed sensing, data collection, and communication in diverse applications ranging from environmental monitoring to industrial automation. Comprising spatially distributed sensor nodes capable of sensing, processing, and communicating, WSNs represent a paradigm shift in how data is captured, processed, and acted upon. The ability of these networks to operate autonomously in dynamic and resource-constrained environments has made them indispensable in addressing complex scientific, industrial, and societal challenges [1].

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This adaptability has led to their deployment in a wide range of domains. In environmental monitoring, WSNs enable predictive systems for floods, landslides, and air pollution through distributed sensing and data fusion techniques. In healthcare, wearable WSNs support continuous physiological monitoring and early detection of anomalies using AI-driven models. Industrial automation leverages WSNs for predictive maintenance, real-time fault detection, and energy management through vibration analysis and hierarchical processing. In precision agriculture, sensor networks guide irrigation and fertilization decisions by analyzing soil moisture, weather conditions, and crop status. Similarly, in disaster response scenarios, WSNs assist in victim localization and structural integrity assessments using acoustic and seismic sensing. These examples demonstrate how WSNs

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enable intelligent decision-making across mission-critical applications [2].

Advancements in microelectronics, communication protocols, and data processing techniques have driven the evolution of WSNs. These advancements have enabled the development of compact, energy-efficient sensor nodes with capabilities far beyond simple data collection. Modern WSNs integrate sophisticated features such as adaptive communication, real-time analytics, and collaborative decision-making, transforming them into intelligent systems beyond passive monitoring to active engagement with their environments. For example, WSNs in smart agriculture not only monitor soil and crop conditions and provide actionable insights to optimize irrigation and fertilization [3].

Despite these advancements, WSNs face significant challenges that limit their scalability, reliability, and operational longevity. Energy constraints are particularly critical, as most sensor nodes rely on batteries with finite capacity. While energy harvesting technologies have provided alternatives, their dependency on environmental conditions introduces variability and uncertainty. Furthermore, the dynamic and often harsh environments in which WSNs operate impose stringent requirements on communication reliability, fault tolerance, and security [4]. Ensuring security and data privacy in WSNs is challenging due to limited computational resources and evolving cyber threats. Lightweight cryptography and blockchain-based authentication enhance protection while maintaining efficiency. Intrusion detection systems (IDS) using machine learning (ML) models improve threat detection and adaptability. Balancing security, energy efficiency, and real-time processing remains a critical challenge.

The rise of the IoT and AI has further expanded the scope of WSNs, introducing new opportunities and challenges. Integration with IoT frameworks has enabled WSNs to become a fundamental component of smart ecosystems. At the same time, AI-driven analytics have enhanced their capability to process and interpret large-scale data streams. However, these integrations also demand greater computational power, robust security mechanisms, and seamless interoperability, necessitating significant advancements in hardware and software design [5].

A. MOTIVATION AND CONTRIBUTION

Table 1 highlights existing surveys on the WSNs domain, and the present survey distinguishes itself by focusing on emerging innovations like quantum sensing, federated intelligence, blockchain-based trust mechanisms, and digital twins integration. Unlike traditional works centred on routing or clustering, this survey emphasizes interdisciplinary relevance by linking WSNs with IoT, AI, and edge computing. It addresses practical challenges such as energy constraints, scalability, and sustainability through future-focused solutions like bio-inspired designs, hybrid communication protocols, and energy management systems. Motivated by gaps in the literature, it explores underrepresented topics like real-time analytics, dynamic topologies, and the application of WSNs in extreme environments, including underwater systems and space exploration. This forward-looking approach positions WSNs as vital for modern technological advancements and intelligent systems. In conclusion, this survey:

- Provides a well-rounded overview covering foundational principles, architectural and energy models, real-world applications, and emerging paradigms of WSNs.
- Provides a structured overview of innovations in WSNs, including advancements in energy harvesting, adaptive communication protocols, edge intelligence, and security mechanisms, emphasizing their impact on modern deployments.
- Systematically identifies and analyzes persistent challenges such as energy constraints, scalability, dynamic topology management, and security vulnerabilities, offering technical insights and design implications for application-specific scenarios.
- Highlights novel research directions, such as quantum sensing, bio-inspired architectures, federated intelligence, integration with IoT ecosystems, software defined WSNs and digital twins, framing their transformative potential for next-generation WSNs.
- By synthesizing fundamentals, applications, innovations, challenges, and future trends, this work is a reference framework to support researchers and practitioners in designing adaptive, resilient, and intelligent WSN infrastructures.

TABLE 1. Overview of existing survey articles in wireless sensor networks.

Reference	Description
[6]	A survey that classifies WSNs deployment techniques, including computational geometry-based, force-based, grid-based, and metaheuristic approaches. It also discusses challenges, performance metrics, and simulators.
[7]	An extensive overview of energy-efficient routing protocols in WSNs, categorized by network structure, data exchange, location use, and quality of service (QoS). Highlights comparative performance metrics and discusses future research directions.
[8]	Analyzes the application of ML for security in WSNs, emphasizing energy-security trade-offs. It identifies challenges like computational limits and proposes ML-based solutions for security across WSN layers.
[9]	A comprehensive survey on clustering in WSNs, focusing on objectives like energy efficiency, load balancing, and QoS. Provides various clustering techniques and their compatibility with network properties.
[10]	Explores localization techniques in mobile WSNs, focusing on random mobility scenarios. Reviews algorithms addressing challenges like computational complexity and mobility unpredictability.
[11]	Surveys low- and high-level design methodologies for WSNs. Discusses the use of model-driven engineering to simplify design and ensure system flexibility and maintainability.
[12]	Discuss mobility-based solutions in WSNs, including mobile nodes and sinks, to enhance network lifetime and throughput. Discusses algorithms for congestion control and topology management.
[13]	Provides an up-to-date analysis of WSN applications across domains like healthcare, environmental monitoring, and military. Explores traditional and novel use cases and highlights their advantages and limitations.

Figure 1 presents a structured overview of the surveyed topics in WSNs: Innovations and Future Trends. It visually organizes the key sections of the paper, illustrating the relationship between foundational WSN components, applications, technological innovations, challenges, and future research directions. The figure highlights key components and technologies, including sensor node architectures, energy models, and communication protocols, which form the backbone of WSN functionality. It further categorizes WSN applications across diverse domains such as environmental monitoring, healthcare, industrial automation, smart cities, precision agriculture, disaster management and emergency response, energy management and renewable energy systems, transportation and logistics, aquatic and underwater monitoring, space exploration and planetary research, unmanned aerial vehicles (UAVs), and military defence, showcasing their impact in real-world deployments. Additionally, the figure outlines technological innovations, including adaptive communication, edge intelligence, energy-efficient designs, AI integration, and security mechanisms, which enhance WSN performance and scalability. Recognizing existing challenges such as energy limitations, network congestion, security threats, and interoperability issues, the diagram sets the stage for future research trends like self-sustaining WSNs, bio-inspired architectures, FL, and quantum sensing. Overall, the figure serves as a conceptual roadmap, guiding readers through the survey's structure while emphasizing the evolution and emerging advancements in WSN technology.

The remaining paper is structured as follows. Section II outlines key components and technologies in WSNs, Section III discusses applications of WSNs, Section IV highlights innovations in WSN technologies, Section V notes challenges in WSN applications, and Section VI explores future trends and research directions. Finally, Section VII concludes the findings of this survey.

II. KEY COMPONENTS AND TECHNOLOGIES IN WIRELESS SENSOR NETWORKS

WSNs operate on an intricate framework of architectures, algorithms, and mathematical techniques, each tailored to optimize system efficiency, reliability, and scalability. This section delves into the fundamental components and technologies that maintain modern WSNs, emphasizing architectural designs, communication methodologies, and parameter estimation techniques.

A. ARCHITECTURAL MODELS IN WSNs

Sensor node architecture is the backbone of WSNs, defining the system's operational capabilities, energy efficiency, and scalability. It comprises multiple interconnected subsystems, including sensing, processing, communication, and power management. Modern sensor node architectures have evolved to address the specific challenges of WSN environments, including resource constraints, network scalability, and operational longevity. This subsection explores the nuances of different architectures (single-tier and multi-tier, centralized, decentralized, and distributed, and duty-cycled), emphasizing their design principles, performance characteristics, and application contexts [14], [15].

1) SINGLE-TIER AND MULTI-TIER ARCHITECTURES

Single-tier architectures represent a foundational approach in WSN design, where each sensor node is identical in functionality and performs all tasks independently. Nodes in this architecture sense environmental data, process it locally, and communicate it to a sink or base station. This uniformity simplifies network design and deployment, making single-tier architectures suitable for small-scale or static applications where uniform sensing is sufficient [16].

However, single-tier architectures often suffer from scalability and energy inefficiency. Nodes closer to the sink bear a higher communication burden due to their role in relaying data from distant nodes, leading to rapid energy depletion in these critical nodes and potentially causing network partitioning [17]. Despite these challenges, single-tier architectures are widely adopted for applications requiring uniform data collection, such as indoor environmental monitoring, where simplicity and cost-effectiveness are prioritized over scalability [18], [19].

Multi-tier architectures introduce a hierarchical structure that mitigates the limitations of single-tier systems by assigning specialized roles to different nodes based on their capabilities and positions. In a typical multi-tier design, lower-tier nodes focus on local data collection and transmission to higher-tier nodes, which aggregate and process the data before forwarding it to the sink. This arrangement significantly reduces communication overhead for lower-tier nodes, extending their operational lifespan and enhancing the overall network efficiency [20], [21].

Multi-tier architectures outperform in scalability, as their hierarchical organization minimizes redundant communication and localizes processing tasks. They also enhance fault tolerance, as higher-tier nodes can dynamically redistribute roles among themselves in the event of node failure. Despite these advantages, the complexity of clustering algorithms, role assignment, and inter-tier synchronization adds to the design and operational overhead. Applications such as smart city infrastructure and industrial automation benefit significantly from multi-tier architectures, as they can support diverse sensing requirements while maintaining energy efficiency and robustness [22], [23].

2) CENTRALIZED, DECENTRALIZED, AND DISTRIBUTED ARCHITECTURES

The architectural paradigm of a WSN plays a decisive role in determining its scalability, reliability, and energy efficiency. Depending on how tasks such as data aggregation, decision-making, and control are distributed among nodes, WSN architectures are broadly categorized into centralized, decentralized, and distributed models. Each architecture has distinct operational characteristics, trade-offs, and suitability

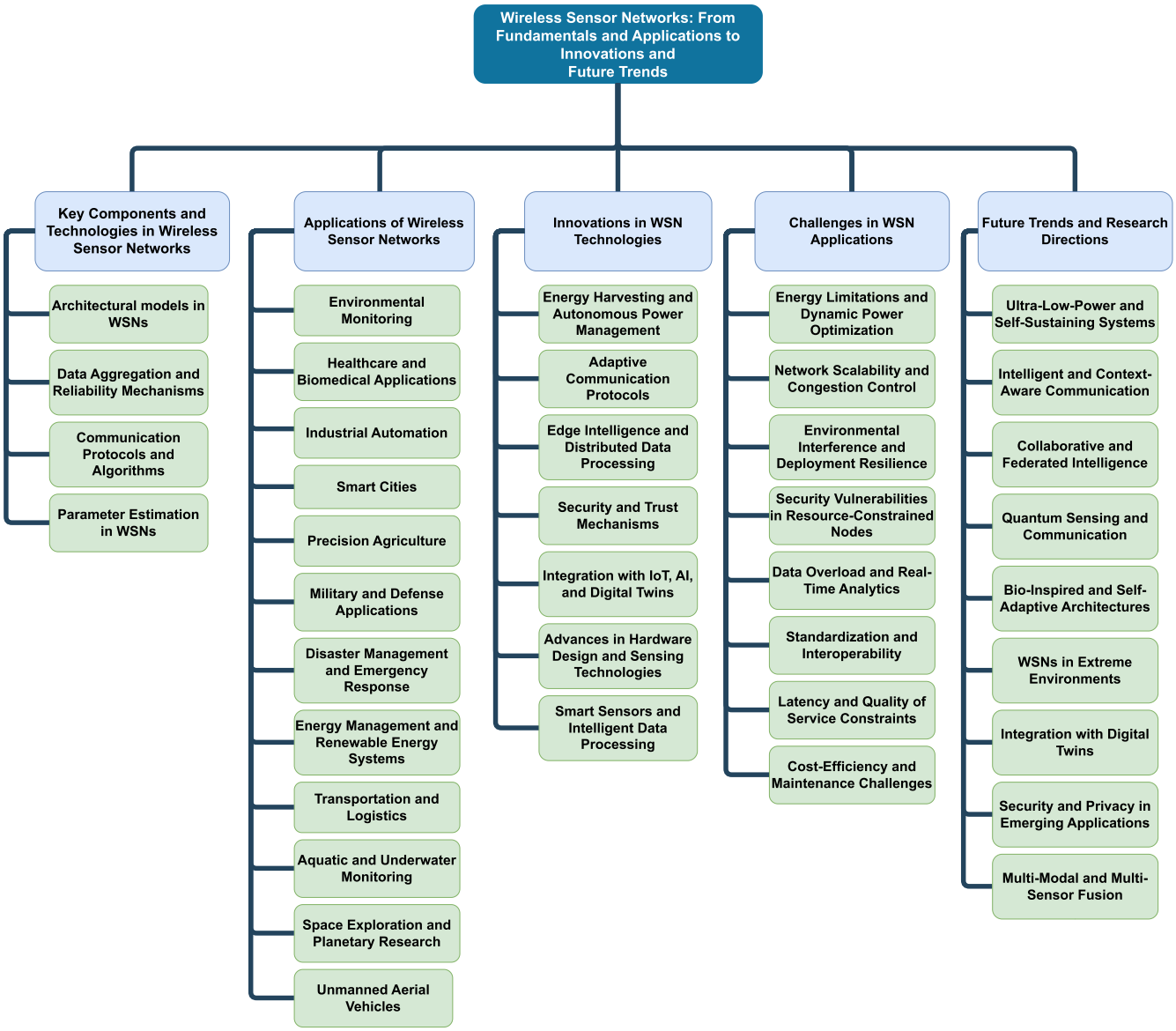


FIGURE 1. Surveyed Key Topics in Applications of WSNs: Innovations and Future Trends.

for specific applications. This subsection provides a focused analysis of these architectures, examining their design principles, mathematical underpinnings, and targeted applications [24].

a: CENTRALIZED ARCHITECTURES

A single control point characterizes centralized architectures, typically a sink or base station, that performs data aggregation, processing, and decision-making for the entire network. In this model, all sensor nodes transmit their collected data directly or indirectly to the sink, which acts as the computational hub. This arrangement simplifies the design and implementation of the network, as the sink has a global view of the data and can execute complex computational tasks that individual sensor nodes cannot perform [25], [26], [27].

The primary advantage of centralized architectures is the ease of implementing sophisticated data analytics and optimization algorithms at the sink. This capability makes them highly effective for applications requiring intensive data processing, such as environmental modeling or industrial diagnostics. However, centralized architectures suffer from significant limitations, including the potential for bottlenecks at the sink and the lack of fault tolerance. Since all nodes transmit directly to a single sink, the failure of that central node can result in complete network disruption. Furthermore, communication range and power requirements increase for nodes located farther from the sink. This can result in uneven energy consumption across the network and early depletion of distant nodes, ultimately limiting the overall network lifetime [28], [29].

Applications of centralized architectures are typically found in small-scale or static deployments where network scalability and redundancy are not primary concerns. For instance, greenhouse monitoring systems often adopt a centralized design, as the spatial constraints and low node density make this approach efficient [30].

b: DECENTRALIZED ARCHITECTURES

Decentralized architectures mitigate the drawbacks of centralized systems by introducing hierarchical structures that distribute computational and communication tasks across multiple nodes. This model organizes sensor nodes into clusters, each managed by a cluster head (CH). The CH aggregates data from its member nodes, processes it locally, and forwards the results to the sink or higher-tier nodes. This hierarchical design reduces the communication overhead and energy consumption for individual sensor nodes, particularly those in lower tiers [31], [32].

Decentralized architectures excel in scalability and fault tolerance. By distributing roles and responsibilities, they prevent the formation of single points of failure and can reconfigure dynamically in response to node failures or changes in network topology. These features make decentralized architectures ideal for applications in smart cities, precision agriculture, and industrial automation, where the network must adapt to variable conditions and maintain consistent performance [33], [34].

Despite their advantages, decentralized architectures introduce challenges related to cluster formation, CH selection, and inter-cluster synchronization. Low-Energy Adaptive Clustering Hierarchy (LEACH) is widely used as it was one of the first protocols to introduce clustering in WSNs. However, it employs a basic clustering mechanism with random CH selection. It lacks advanced features such as adaptive decision-making, multi-hop communication, and energy-aware optimization, making it less efficient than more sophisticated approaches (detailed in the routing protocols section) that balance energy efficiency with network performance [35].

While decentralized and multi-tier architectures originate from different perspectives (namely, protocol-level organization versus system-level hierarchy), they are structurally aligned. Both arrange sensor nodes into hierarchical layers, where lower-tier nodes communicate with coordinating entities termed CHs in decentralized systems and higher-tier nodes in multi-tier designs. These coordinators handle data aggregation and forwarding, reducing communication overhead and enhancing scalability. Despite the varied terminology, the roles and communication patterns are functionally equivalent, allowing for unified modeling in energy and performance analysis.

c: DISTRIBUTED ARCHITECTURES

Distributed architectures represent the most decentralized form of WSN design, where all sensor nodes collaboratively participate in data aggregation, processing, and decision-making without relying on a single control point or hierarchical structure. Each node operates autonomously, making local decisions based on observations and interactions with neighbouring nodes. This peer-to-peer design offers unparalleled scalability and fault tolerance, as the failure of individual nodes does not compromise the network's overall functionality [36], [37].

The distributed nature of this architecture relies heavily on consensus algorithms to synchronize decisions across the network. A typical example is the average consensus algorithm, which computes a global parameter, such as the average temperature, based on local observations. The iterative update rule for the state x_i of node i at iteration $k+1$ is given by $x_i^{(k+1)} = x_i^{(k)} + \epsilon \sum_{j \in \mathcal{N}_i} \left(x_j^{(k)} - x_i^{(k)} \right)$, where $\mathcal{N}_i$ represents the neighbors of node i, and ϵ is the step size that ensures stability and convergence [38], [39], [40].

The energy consumption in distributed architectures is distributed more evenly across nodes, as all nodes share the computational and communication workload. However, this egalitarian structure also means that distributed architectures require robust coordination protocols to prevent inconsistencies and ensure efficient resource utilization [41], [42].

Distributed architectures are particularly suited for dynamic and large-scale networks, such as environmental monitoring over vast regions or disaster response scenarios. Their ability to adapt to rapidly changing conditions and operate autonomously without centralized control is invaluable in such contexts [43].

3) DUTY-CYCLED ARCHITECTURES

Duty-cycled architectures tackle the fundamental challenge of energy efficiency in WSNs by periodically transitioning nodes between active and sleep states. In this paradigm, energy consumption is minimized by limiting node activity to essential operations such as sensing, computation, and communication, while maximizing low-power sleep durations during idle periods. The operation of a duty-cycled node is characterized by the duty cycle D, which quantifies the fraction of time the node remains active within a given operational period $D = \frac{T_{\text{active}}}{T_{\text{cycle}}}$, where T_{active} is the duration of the active phase, and T_{cycle} refers to the total duration of one complete cycle. This operational cycle consists of two phases: the active phase (T_{active}) and the sleep phase (T_{sleep}). The full cycle duration is therefore given by $T_{\text{cycle}} = T_{\text{active}} + T_{\text{sleep}}$. This simple yet effective strategy significantly extends network lifetime, especially when continuous monitoring is not required [44], [45], [46].

Advanced duty-cycled architectures dynamically adjust T_{active} and T_{sleep} based on the network's operational demands. For instance, nodes in regions with high event activity may increase their duty cycles to ensure timely data reporting. In contrast, nodes in less active areas reduce their duty cycles to conserve energy.

Duty-cycled architectures are particularly suited for applications requiring long-term monitoring in energy-constrained environments. Examples include environmental sensing for climate studies or wildlife tracking, where conserving energy is critical for extending network longevity. Despite their benefits, duty-cycled architectures require precise synchronization among nodes to maintain network connectivity and data consistency, which can be challenging in large-scale or dynamic deployments [47], [48].

Table 2 provides a comparative overview of key WSN architectures. It contrasts single-tier, multi-tier, and duty-cycled architectures, highlighting scalability, energy efficiency, and adaptability trade-offs. While single-tier architectures are simple but less scalable, multi-tier models improve load distribution, and duty-cycled systems enhance energy conservation. The table also compares centralized, decentralized, and distributed architectures, where centralized models face single points of failure, while decentralized and distributed designs improve fault tolerance and scalability.

B. DATA AGGREGATION AND RELIABILITY MECHANISMS

Data aggregation techniques reduce communication overhead by combining data from multiple sensor nodes before transmission, minimizing redundancy and improving energy efficiency. This process is particularly beneficial in environmental monitoring, precision agriculture, and structural health monitoring applications, where distributed sensors generate large volumes of similar or correlated data. By aggregating data at intermediate nodes, the packets transmitted across the network are significantly reduced, leading to lower energy consumption and extended network lifetime.

Several aggregation techniques have been proposed to optimize data fusion and minimize redundant transmissions. Hierarchical aggregation organizes nodes into clusters where a designated CH collects and processes data before forwarding it to the sink. Tree-based aggregation constructs a hierarchical routing tree where intermediate nodes perform iterative aggregation as data is relayed toward the sink, reducing redundant transmissions [49]. Adaptive aggregation mechanisms adjust the aggregation level dynamically based on network conditions, ensuring trade-offs between accuracy and efficiency.

Despite its benefits, data aggregation introduces challenges such as increased processing delay, data integrity risks, and compromised security, as aggregating nodes may become targets for adversarial attacks. Secure aggregation techniques [50] leveraging homomorphic encryption, lightweight and cryptographic methods have been developed to mitigate these risks while preserving efficiency.

Error control mechanisms ensure data integrity during transmission by addressing issues such as interference and packet loss. Forward Error Correction (FEC) adds redundant data to packets, enabling receivers to correct certain errors without requiring retransmission [51]. The trade-off involves increased computational energy for encoding and decoding. Alternatively, Automatic Repeat Request (ARQ) relies on retransmitting corrupted packets, increasing reliability at the expense of additional communication energy [52], [53]. The energy trade-off can be expressed as $E_{\text{error}} = E_{\text{transmit}} + E_{\text{process}} \cdot (1 + r)$, where r represents the redundancy factor.

Time synchronization is critical for coordinating node operations in WSNs, especially in duty-cycled networks. Algorithms such as Reference Broadcast Synchronization (RBS) achieve synchronization through broadcast messages. The synchronization error e_s is minimized iteratively as $e_s^{(k+1)} = \alpha e_s^{(k)} + \beta$, where α and β are constants that depend on the system's dynamics [54], [55].

By integrating data aggregation, error control, and time synchronization, WSNs achieve improved energy efficiency, reduced communication overhead, and enhanced reliability. These mechanisms are fundamental to optimizing network operation, particularly in energy-constrained and large-scale deployments. Ongoing research continues to refine these techniques by incorporating ML-driven aggregation strategies, adaptive error correction, and AI-based time synchronization, further enhancing WSN performance in dynamic environments.

C. ARCHITECTURAL ENERGY MODELING IN WSNs

Energy efficiency is a fundamental design objective in WSNs, where the operational lifetime of sensor nodes is directly constrained by limited battery capacity. Independent of the architectural paradigm (e.g., centralized, hierarchical, or duty-cycled), accurate energy consumption modeling is essential for evaluating protocol behavior, designing energy-aware mechanisms, and extending network longevity.

The energy consumed during a complete operational cycle can be expressed through a baseline model that accounts for all major energy-consuming operations, including sensing, processing, transmitting, and receiving

$$E_{\text{baseline}} = \sum_{i=1}^{N} \left(E_{\text{sense},i} + E_{\text{process},i} + E_{\text{transmit},i} + E_{\text{receive},i} \right), \tag{1}$$

where N is the number of sensor nodes in the network, and $E_{\text{sense},i}$, $E_{\text{process},i}$, $E_{\text{transmit},i}$, and $E_{\text{receive},i}$ represent the energy consumed by node i during sensing, in-node processing, data transmission, and data reception, respectively.

For centralized architectures, where each node transmits its sensed data directly to the sink without intermediate processing or multi-hop relaying, the total energy consumption [28], [29] to

$$E_{\text{centralized}} = \sum_{i=1}^{N} \left(E_{\text{sense},i} + E_{\text{transmit},i} \right), \tag{2}$$

where $E_{\text{sense},i}$ and $E_{\text{transmit},i}$ represent the energy consumed by node i for sensing and direct data transmission to the sink, respectively. This expression derives from the baseline model

TABLE 2. Comparative analysis of WSN architectures.

Aspect	Single-Tier	Multi-Tier	Duty-Cycled	Centralized	Decentralized	Distributed
Key Features	Uniform nodes; common in basic systems like indoor temperature monitoring.	Hierarchical design with distinct roles for lower- and higher-tier nodes; e.g., smart city applications.	Nodes periodically alternate between active and sleep states to conserve energy; ideal for wildlife monitoring.	Single sink node aggregates all data and makes decisions; suitable for small, static deployments.	CHs manage local processing and aggregation; often used in precision agriculture.	Fully autonomous nodes collaborate for decision-making; used in disaster response or robotic coordination.
Scalability	Limited scalability due to bottlenecks in communication and energy depletion near the sink.	Highly scalable as CHs manage communication load locally.	Moderate scalability depends on synchronization and duty-cycle management.	Limited scalability due to single-point bottlenecks at the sink.	High scalability as clusters distribute workloads efficiently.	Very high scalability through autonomous node collaboration and dynamic adjustments.
Energy Efficiency	Low; High energy drain near the sink.	Moderate; CHs consume more energy but reduce redundant transmissions.	High efficiency by reducing idle listening; energy conservation depends on synchronization accuracy.	Poor efficiency for nodes near the sink due to heavy relay loads.	Moderate; energy balanced across clusters; longer lifetime.	Very high efficiency due to local data processing and reduced long-range communication.
Reliability	Vulnerable to failures near the sink due to centralized communication.	Fault-tolerant at the cluster level; dynamic cluster reconfiguration improves reliability.	Highly reliable when synchronization is accurate; failures in timing degrade performance.	Prone to single-point failures at the sink; reliability decreases with sink overload.	Improved reliability via cluster-level redundancy and fault tolerance.	Highly reliable with dynamic reconfiguration and multi-path routing.
Real-Time Adaptability	Limited adaptability to changing environments.	Moderate adaptability with re-clustering and dynamic role allocation.	High adaptability; duty cycles can be adjusted dynamically based on environmental conditions.	Limited adaptability; sink-driven decision-making causes delays in dynamic environments.	Moderate adaptability through hierarchical adjustments at cluster levels.	Very high adaptability using consensus algorithms and iterative updates; suitable for dynamic networks.
Computational Complexity	Minimal; relies on sink for processing and decision-making.	Moderate; CHs handle aggregation and processing tasks.	Low complexity at the node level; energy-efficiency algorithms add minimal overhead.	Centralized computation reduces node load but burdens the sink.	Moderate complexity due to cluster synchronization and inter-cluster communication.	High complexity due to distributed consensus, coordination, and autonomous operation.

E_{baseline} by omitting the reception and processing terms, which are negligible in one-hop flat topologies.

In hierarchical WSNs, such as multi-tier or decentralized-cluster-based architectures, energy consumption is distributed across regular sensor nodes and CHs that coordinate intra- and inter-cluster communication. These architectures typically rely on a local communication phase, where sensor nodes transmit their sensed data to the nearest CH, which then performs data aggregation and forwards the processed information to the sink. The total energy consumption for such systems can be modeled as

$$E_{\text{clustered}} = \sum_{j=1}^{C} \left(\sum_{i \in N_j} E_{\text{local},i} \right) + \sum_{j=1}^{C} E_{\text{CH},j}, \tag{3}$$

where C is the number of clusters, N_j is the set of nodes assigned to cluster j, $E_{\text{local},i}$ denotes the energy consumed by sensor node i for sensing, processing, and intra-cluster communication, and $E_{\text{CH},j}$ represents the energy consumed by the CH of cluster j for data aggregation and inter-cluster communication to the sink [56], [57].

This model reflects the layered structure of clustered WSNs and enables comparative analysis against baseline and centralized architectures. The inclusion of CHs significantly reduces long-range transmissions for non-CH nodes, thereby improving overall energy efficiency, though it also increases the energy burden at the cluster-head level.

In multi-tier and decentralized WSN architectures, data aggregation is commonly employed at intermediate nodes to reduce redundant transmissions and alleviate communication overhead. This mechanism improves energy efficiency by shifting part of the transmission burden to localized processing. The corresponding energy cost can be modeled as

$$E_{\text{aggregation}} = E_{\text{process,CH}} + \sum_{i \in P} \left(E_{\text{transmit},i} + E_{\text{receive},i} \right), \tag{4}$$

where $E_{\text{process,CH}}$ denotes the energy consumed at designated aggregation nodes (e.g., CHs) for data fusion, compression,

or filtering, and $E_{\text{transmit},i}$, $E_{\text{receive},i}$ represent the communication energy costs incurred along the forwarding path P [58], [59], [60]. This formulation complements the architectural models discussed previously by explicitly accounting for the functional overhead of in-network processing.

While the clustered architecture model captures energy expenditure as the sum of local node costs and cluster-head contributions ($E_{\text{CH},j}$), it does not explicitly decompose the internal energy components of the CH. The aggregation model refines this by isolating the computational energy for data fusion ($E_{\text{process,CH}}$) and the communication load carried by the nodes relaying the aggregated data to the sink. This added granularity is particularly valuable for analyzing trade-offs between local computation and transmission reduction, everyday design decisions in multi-tier and decentralized WSN deployments. As such, the aggregation model functions as a complementary layer within the broader architectural energy modeling framework, enabling more precise evaluation of energy-aware protocol design and role allocation.

In duty-cycled WSN architectures, sensor nodes periodically alternate between active and sleep states to conserve energy. The average energy consumption in such systems can be approximated by

$$E_{\text{duty-cycle}} = \sum_{i=1}^{N} \left(D_i \cdot P_{\text{active},i} + (1 - D_i) \cdot P_{\text{sleep},i} \right), \quad (5)$$

where D_i represents the duty cycle (i.e., the fraction of time spent in the active state) for node i, and $P_{\text{active},i}$, $P_{\text{sleep},i}$ denote the corresponding power levels during active and sleep modes, respectively [61], [62].

This formulation captures the dynamics of energy usage and complements the previously presented architectural models. Collectively, the proposed modeling framework offers a comprehensive basis for evaluating the energy efficiency of various WSN deployment strategies. It also establishes a foundation for the optimization methodologies introduced in the following sections.

D. COMMUNICATION PROTOCOLS AND ALGORITHMS

Communication protocols and algorithms are fundamental to ensuring efficient and reliable data exchange in WSNs. These methods are designed to address the unique challenges of WSNs, including energy constraints, dynamic topology, and limited computational resources. This subsection explores Medium Access Control (MAC) and routing protocols, highlighting key operational principles [63].

1) MEDIUM ACCESS CONTROL PROTOCOLS

MAC protocols manage access to the shared communication medium, ensuring collision-free and energy-efficient data transmission. These protocols are vital in reducing idle listening and minimizing energy consumption. At a high level, they are classified into Contention-based, Contention-free (Synchronous, Asynchronous), and Hybrid.

a: CONTENTION-FREE

Several contention-free MAC protocols have been developed to enhance energy efficiency, reliability, and scalability in WSNs by eliminating collisions and minimizing retransmissions. Time Division Multiple Access (TDMA) assigns dedicated time slots to nodes, ensuring collision-free communication and reducing energy waste, making it particularly effective in clustered networks such as LEACH. Similarly, Frequency Division Multiple Access (FDMA) provides an alternative by allocating different frequency bands to nodes, allowing simultaneous transmissions while minimizing interference. Unlike carrier-sense multiple access with collision avoidance (CSMA/CA), which relies on a shared channel and is prone to congestion and collisions, FDMA improves channel utilization and reduces retransmissions. However, its implementation requires precise frequency synchronization and more complex hardware, making it less suitable for energy-constrained WSNs than TDMA and hybrid approaches. Despite these challenges, FDMA is advantageous in applications that demand continuous, real-time data transmission, such as industrial automation and environmental monitoring, where dedicated frequency channels enhance reliability and throughput. In modern WSNs, FDMA is often combined with TDMA or Code Division Multiple Access (CDMA) to improve scalability and efficiency, particularly in large-scale deployments. Finally, among modern contention-free protocols, IEEE 802.15.4e TSCH (Time Slotted Channel Hopping) enhances reliability and mitigates interference through synchronized time slots and frequency [64].

b: CONTENTION-BASED

Dense WSNs often rely on contention-based protocols such as CSMA/CA. However, these protocols face challenges like higher collision rates, particularly in high-density scenarios. To mitigate this, mechanisms such as RTS/CTS (Request to Send/Clear to Send) handshaking are employed to reduce packet collisions and improve transmission reliability [65], [66].

Contention-based MAC protocols can be broadly classified into synchronous and asynchronous approaches. Synchronous MAC protocols, such as Sensor-MAC (S-MAC) and Timeout-MAC (T-MAC), employ periodic sleep/wake cycles to conserve energy. T-MAC optimizes energy efficiency by dynamically adjusting active periods based on network traffic conditions, reducing latency during high-traffic periods at the cost of slightly increased energy consumption. Also, dynamic MAC (DMAC) adjusts duty cycling based on traffic patterns, ensuring adaptive power management [67], [68]. In synchronous MAC protocols, nodes alternate between active and sleep states in synchronized cycles, where the frame time T_{frame} is given by $T_{\text{frame}} = T_{\text{listen}} + T_{\text{sleep}}$, where T_{listen} and T_{sleep} denote the listening and sleeping durations, respectively [69], [70].

Unlike synchronous approaches, asynchronous MAC protocols operate without global synchronization, allow-

ing nodes to follow independent schedules. This enables flexible duty cycling, reducing idle listening and energy waste. WiseMAC, for example, minimizes idle listening by allowing the nodes to wake up only when a transmission is expected, using preamble sampling. Several asynchronous MAC protocols have been developed to optimize energy efficiency further. X-MAC is designed as an enhancement over Berkeley MAC (B-MAC), improving preamble handling by shortening preambles and enabling early acknowledgment. Receiver-Initiated MAC (RI-MAC) shifts transmission initiation to the receiver, reducing idle listening. B-MAC employs Low-Power Listening (LPL) and adaptive preamble sampling to balance energy efficiency and latency. These and other asynchronous protocols enhance adaptability and energy conservation in low-duty-cycle WSN applications by reducing unnecessary transmissions and improving medium access efficiency [71].

c: HYBRID

Hybrid protocols, such as Zebra-MAC (Z-MAC), adaptively switch between contention-based access under low traffic and TDMA-based scheduling when congestion increases, improving efficiency. More advanced approaches integrate cross-layer optimization to improve performance further.

d: ADAPTIVE

Recently, Reinforcement Learning (RL)-based MAC protocols [72] have gained attention for their ability to dynamically learn optimal transmission strategies by adapting to changing network conditions. These approaches leverage deep RL techniques to optimize channel access, minimize collisions, and improve energy efficiency without relying on predefined scheduling rules. By leveraging these alternative MAC protocols, WSNs can achieve better energy conservation, reduced latency, and increased robustness, depending on application-specific requirements.

A comprehensive classification of key MAC protocols used in WSNs is presented in Table 3, detailing their access methods and key features. In addition to architectural and functional characteristics, the table integrates representative energy consumption models for each protocol class, emphasizing the operational parameters involved, specifically the time duration in different states ($T_{\text{transmit}}, T_{\text{receive}}, T_{\text{listen}}, T_{\text{sleep}}, T_{\text{preamble}}, T_{\text{idle}}, T_{\text{collisions}}$) and their corresponding power levels ($P_{\text{transmit}}, P_{\text{receive}}, P_{\text{listen}}, P_{\text{sleep}}, P_{\text{preamble}}, P_{\text{idle}}, P_{\text{collisions}}$). This detailed integration enables direct comparison of MAC protocols not only in terms of access strategies and design philosophy but also concerning their quantitative impact on node-level energy efficiency.

2) ROUTING PROTOCOLS

Routing protocols in WSNs play a crucial role in determining efficient paths for data transmission from sensor nodes to the sink while considering energy constraints, scalability, mobility, and security. Selecting a suitable routing protocol depends on the network topology, communication requirements, and operational constraints. Various routing protocols have been proposed. These protocols are designed to minimize energy consumption, ensure reliability, and balance traffic loads [74]. The following paragraphs are categorized based on their key objectives and techniques.

a: DATA-CENTRIC

Data-centric routing protocols enhance energy efficiency in WSNs by minimizing redundant transmissions through in-network aggregation and interest-driven forwarding. Unlike traditional address-centric approaches, they employ a query-response mechanism to retrieve relevant data, dynamically adjusting transmission paths and eliminating unnecessary messages. These capabilities make data-centric protocols particularly effective for applications requiring prolonged network operation under energy constraints.

One of the fundamental characteristics of data-centric routing is its ability to employ interest propagation and data aggregation to optimize communication efficiency. The sink node broadcasts an interest query to initiate data collection. Nodes that contain relevant information respond by forwarding their data along optimized paths, where intermediate nodes aggregate similar data before relaying it further. This aggregation process minimizes redundant packet transmissions, reduces congestion, and conserves energy. The total energy consumption in data-centric routing protocols can be expressed as

$$E_{\text{data-centric}} = E_{\text{query}} + E_{\text{forwarding}} + E_{\text{aggregation}} + E_{\text{data-transmission}} \quad (6)$$

where E_{query} represents the energy used for interest dissemination, $E_{\text{forwarding}}$ corresponds to the energy used for setting up data gradients, $E_{\text{aggregation}}$ accounts for the energy utilized in compressing and aggregating data at intermediate nodes, and $E_{\text{data-transmission}}$ is the actual energy used for sending aggregated data to the sink [75], [76].

Directed Diffusion (DD) [77] is a well-known example that employs reinforcement-based path selection. It propagates interest messages from the sink node to establish data gradients, guiding packets through optimized paths. Similarly, Rumor Routing (RR) minimizes query flooding by deploying long-lived agents that traverse the network to disseminate event information. Another approach, COUGAR, introduces a query-processing layer that integrates data from multiple sources, optimizing the overall communication overhead.

In addition to Rumor and COUGAR, several other protocols enhance data dissemination and adaptability in WSNs. Sensor Protocols for Information via Negotiation (SPIN) minimizes redundant transmissions through negotiation, while Gradient-Based Routing (GBR) employs hop-based gradient routing for efficient data forwarding. Moreover, Constrained Anisotropic Diffusion Routing (CADR) dynamically adjusts data propagation to balance network load,

TABLE 3. Classification of MAC protocols in WSNs and associated energy models.

Category	Protocol	Access Method	Key Features	Energy Consumption Model
Contention-Free MAC Protocols				
Schedule-Based	TDMA	Time-slotted channel access	Assigns dedicated time slots to nodes, preventing collisions but requiring strict synchronization	$E_{\text{TDMA}} = \sum_{i=1}^{N}(T_{\text{transmit},i}P_{\text{transmit},i} + T_{\text{receive},i}P_{\text{receive},i} + T_{\text{sleep},i}P_{\text{sleep},i})$ where $T_{\text{sleep},i}$ accounts for the sleep duration of nodes, which helps reduce energy consumption.
	FDMA	Frequency-separated channel access	Allocates different frequency bands to nodes, reducing interference but increasing hardware complexity	$E_{\text{FDMA}} = \sum_{i=1}^{N}(T_{\text{transmit},i}P_{\text{transmit},i} + T_{\text{receive},i}P_{\text{receive},i} + T_{\text{idle},i}P_{\text{idle},i})$ $T_{\text{idle},i}$ represents idle listening periods when no data is being transmitted or received
	CDMA	Code-separated channel access	Uses unique codes for node transmissions, enabling simultaneous communication with reduced interference	Modeled similarly to FDMA but adds code processing overhead (not explicitly modeled)
	IEEE 802.15.4e TSCH	Time-slotted and frequency-hopping access	Synchronizes time slots and hop frequencies to enhance reliability and mitigate interference	Follows TDMA model with additional frequency switching and synchronization maintenance costs.
Contention-Based MAC Protocols				
Synchronous	S-MAC	Periodic sleep/wake cycles	Implements fixed duty cycles to reduce energy consumption, though it increases latency	$E_{\text{S-MAC}} = \sum_{i=1}^{N}(T_{\text{listen},i}P_{\text{listen},i} + T_{\text{transmit},i}P_{\text{transmit},i} + T_{\text{sleep},i}P_{\text{sleep},i})$
	T-MAC	Adaptive sleep scheduling	Dynamically adjusts wake-up times based on network traffic, reducing idle listening	Same as S-MAC with variable T_{sleep} and T_{listen}
	DMAC	Traffic-aware duty cycling	Adapts duty cycles to transmission patterns, balancing energy efficiency and responsiveness	Similar to S-MAC, optimized for latency-aware scheduling
Asynchronous	CSMA/CA	Carrier sensing with collision avoidance	Nodes sense the channel before transmission and use RTS/CTS to reduce collisions	$E_{\text{CSMA/CA}} = \sum_{i=1}^{N}(T_{\text{listen},i}P_{\text{listen},i} + T_{\text{transmit},i}P_{\text{transmit},i} + T_{\text{receive},i}P_{\text{receive},i} + T_{\text{collisions},i}P_{\text{collisions},i})$ $T_{\text{listen},i}$: time spent in listening, $T_{\text{transmit},i}$: time spent in transmiting, $T_{\text{receive},i}$: time spent in receiving, and $T_{\text{collisions},i}$: time spent in handling collisions
	B-MAC	LPL	Uses preamble sampling to minimize idle listening and conserve energy	$E_{\text{Async}} = \sum_{i=1}^{N}(T_{\text{preamble},i}P_{\text{preamble},i} + T_{\text{listen},i}P_{\text{listen},i} + T_{\text{transmit},i}P_{\text{transmit},i} + T_{\text{sleep},i}P_{\text{sleep},i})$
	X-MAC	Short preamble transmission	Uses short preamble bursts instead of long preambles to improve energy efficiency and reduce delay	Follows E_{Async} model with smaller T_{preamble}
	RI-MAC	Receiver-initiated communication	Receiver signals readiness before transmission, reducing idle listening	Variant of E_{Async} model optimized for reduced idle listening
	WiseMAC	Preamble-based wake-up scheduling	Predicts wake-up times to optimize transmission	Same as E_{Async} with preamble scheduling
Hybrid and Adaptive MAC Protocols				
Hybrid	Z-MAC	Adaptive contention and scheduling	Uses CSMA under low traffic and switches to TDMA in high-traffic scenarios	$E_{\text{Hybrid}} = \sum_{i=1}^{N}(\alpha \cdot E_{\text{CSMA},i} + (1-\alpha) \cdot E_{\text{TDMA},i})$ α is a traffic-dependent weighting factor that determines the protocol's mode of operation [73]
Adaptive	RL-MAC	AI-driven channel access optimization	Uses RL to dynamically adjust transmission strategies based on network conditions	Learning-based adaptation of E_{MAC} models with real-time updates

whereas Two-Tier Data Dissemination (TTDD) enables efficient data retrieval in networks with mobile sinks [80]. In addition to the core data-centric protocols, approaches such as Energy-Aware Routing (EAR) selects energy-efficient

paths, which prevents node depletion [78] and Event-to-Sink Reliable Transport (ESRT), which optimizes event reliability with adaptive rate control [79], also contribute to the broader data-centric paradigm, despite not being strictly data-centric in design.

These protocols collectively enhance network longevity and communication efficiency across various applications. However, they are not the only data-centric routing approaches proposed for WSNs. Numerous other protocols have been developed, each tailored to specific challenges such as scalability, mobility, fault tolerance, and security. Variants incorporating ML, RL, and adaptive optimization continue to emerge, further refining data dissemination strategies to meet the evolving demands of modern sensor networks.

b: HIERARCHICAL (CLUSTER-BASED)

Hierarchical routing protocols enhance energy efficiency by organizing nodes into clusters, where a designated CH is responsible for aggregating and transmitting data to the sink. This approach reduces the number of direct transmissions to the sink, significantly conserving energy. The formula for capturing the total energy consumption in a cluster-based routing protocol is captured by (3).

Several hierarchical routing protocols have been developed, each focusing on optimizing energy consumption and network longevity. LEACH [81], [82] is one of the most widely used protocols, periodically selecting new CHs to distribute energy consumption across the network evenly. Power Efficient Adaptive Clustering Hierarchy (PEACH) enables adaptive clustering formation without requiring explicit control messages, reducing overhead and improving energy efficiency. Unlike traditional clustering schemes, PEACH leverages overheard packets to dynamically form clusters, making it particularly effective in environments with unpredictable topology changes. Power-Efficient GAthering in Sensor Information System (PEGASIS) organizes sensor nodes into a chain-based topology, where each node transmits data to its nearest neighbor in a sequential manner until reaching a designated leader node, which aggregates the collected data and transmits it to the sink. This approach reduces the number of direct transmissions to the sink, minimizes long-distance communication, significantly lowers energy consumption, and balances network lifetime by distributing the communication load among nodes over multiple rounds. Moreover, threshold-sensitive Energy Efficient Network (TEEN) is tailored for time-critical applications by introducing hard and soft thresholds, which control event-driven data transmission. Adaptive TEEN (APTEEN) extends TEEN by incorporating both periodic and event-driven communication, making it more flexible for diverse applications [83]. Meanwhile, Hybrid Energy-Efficient Distributed Clustering (HEED) improves CH selection by considering residual energy and node proximity, ensuring a balanced energy distribution and prolonged network lifetime [84].

These protocols continue to evolve, addressing the challenges of scalability, mobility, and energy constraints in large-scale WSN deployments. However, the literature is not limited to these approaches, as numerous other hierarchical routing protocols have been proposed to enhance network efficiency under diverse operational conditions further.

The protocols classified under the data-centric and hierarchical categories, along with their routing mechanisms and key characteristics, are overviewed in Table 4.

c: LOCATION-BASED ROUTING

Location-based or geographic routing protocols utilize location information, obtained through GPS or virtual coordinates, to optimize forwarding decisions, thereby minimizing the number of hops and reducing energy consumption. The efficiency of geographic routing is reflected in its hop-count-based energy consumption model, expressed as

$$E_{\text{geographic}} = \sum_{k=1}^{H} (E_{\text{transmit},k} + E_{\text{receive},k}), \tag{7}$$

where H denotes the total number of hops in the routing path.

Greedy Perimeter Stateless Routing (GPSR) employs a greedy forwarding strategy, transmitting packets to the closest neighbor toward the destination. When a direct route is unavailable, perimeter routing is applied to navigate around obstacles [85]. To further enhance performance, Geographic and Energy Aware Routing (GEAR) refines data forwarding by incorporating both geographic distance and residual energy levels, ensuring balanced energy utilization across nodes. The Self-Organizing Protocol for Wireless Networks (SPAN) introduces an adaptive duty cycling mechanism, where nodes activate dynamically only when required for routing, thereby reducing unnecessary energy expenditure and prolonging network lifetime. Other notable geographic routing protocols include Geographic Random Forwarding (GERAF), Geographic Opportunistic Adaptive Face Routing (GOAFR), Beacon Vector-Based Forwarding (BVGF), and Trajectory-Based Forwarding (TBF), each designed to improve routing resilience in dynamic and large-scale WSN environments [86].

d: MULTIPATH

By distributing traffic between source and sink nodes across multiple routes, multipath protocols mitigate network congestion and enhance overall stability, making them well-suited for large-scale WSN deployments. This redundancy ensures data delivery even in the presence of link failures while optimizing energy consumption across the network. The total energy consumption model for multipath routing is expressed as

$$E_{\text{multipath}} = \sum_{p=1}^{P} \sum_{k=1}^{H_p} (E_{\text{transmit},k} + E_{\text{receive},k}), \tag{8}$$

where P represents the total number of available paths, and H_p denotes the number of hops in path p. By distribut-

TABLE 4. Classification of routing protocols in WSNs.

Category	Protocol	Routing Approach	Key Features
Data-centric	DD	Interest-based gradient routing	Establishes data gradients using query-response mechanisms to minimize redundant transmissions
	RR	Event-driven probabilistic routing	Uses long-lived agents to reduce query flooding and disseminate event information efficiently
	COUGAR	Query-processing layer	Incorporates a database abstraction to optimize in-network query processing
	SPIN	Negotiation-based data dissemination	Reduces redundant transmissions by allowing nodes to negotiate before sending data
	GBR	Hop-based gradient routing	Uses hop-count-based gradients to forward data efficiently
	CADR	Adaptive data propagation	Dynamically adjusts data forwarding to balance network load
	TTDD	2-tier structure for data dissemination	Efficient data retrieval in mobile sink networks
Hierarchical	LEACH	Cluster-based adaptive routing	Periodically rotates CHs to evenly distribute energy consumption
	PEACH	Adaptive clustering without explicit control messages	Forms clusters dynamically based on overheard packets, reducing communication overhead and improving energy efficiency in dynamic network topologies
	PEGASIS	Chain-based hierarchical routing	Forms a chain topology where nodes forward data sequentially to minimize energy use
	TEEN	Threshold-sensitive event-driven routing	Sets hard and soft thresholds to optimize event-based data transmission
	HEED	Hybrid energy-efficient clustering	Dynamically selects CHs based on residual energy and node proximity
	APTEEN	Hybrid event-periodic routing	Extends TEEN by incorporating both periodic and event-driven communication

ing traffic across multiple routes, these protocols reduce congestion, prevent energy depletion in critical nodes, and improve overall network performance. Various techniques have been proposed to optimize energy efficiency and route selection, including energy-aware forwarding and adaptive path selection mechanisms [87].

Multipath routing techniques vary depending on network topology, application requirements, and constraints, with numerous approaches introduced to optimize performance under different operational conditions further. Notable protocols in this category include Multipath Directed Diffusion (MPDD), which extends Directed Diffusion by maintaining multiple paths for increased reliability, Reliable Information Forwarding using Multiple Paths (ReInForM), which prioritizes packets dynamically based on their criticality, and Multipath Routing with Erasure Coding (MREC), which improves resilience by applying erasure coding techniques to recover lost data. These protocols demonstrate the diverse strategies employed in multipath routing to enhance data delivery, energy efficiency, and fault tolerance in WSNs. These protocols, along with several other multipath routing approaches, are discussed in [86] and [88].

e: QoS-AWARE

QoS-aware routing protocols are designed to ensure efficient communication while satisfying QoS requirements such as latency, bandwidth, and reliability. These protocols dynamically adjust routing decisions to optimize network performance based on real-time conditions. The total energy consumption in QoS-aware routing can be expressed as

$$E_{\text{QoS}} = \sum_{k=1}^{H} \left(E_{\text{transmit},k} + E_{\text{receive},k} + E_{\text{error-control},k} \right), \quad (9)$$

where H represents the number of hops, and $E_{\text{error-control},k}$ accounts for additional processing overhead related to error correction and retransmissions. These protocols improve network efficiency by integrating QoS-aware mechanisms, ensuring reliable data delivery while effectively managing energy constraints [89].

Several protocols have been developed in this category to optimize QoS in WSNs. Stateless Protocol for End-to-End Delay (SPEED) employs a stateless geographic-based approach, adjusting forwarding decisions according to estimated transmission delays to guarantee real-time packet delivery. Sequential Assignment Routing (SAR) incorporates multiple QoS metrics, including priority levels and energy constraints, to enhance data delivery reliability, particularly in mission-critical applications. Energy-Efficient QoS-Aware Routing (EEQR) balances energy efficiency with delay constraints, ensuring sustained network operation by dynamically selecting optimal paths. Minimum Cost Forwarding Algorithm (MCFA) assigns cost values to paths based on energy and delay metrics, enabling efficient route selection while minimizing resource consumption. Energy-Aware QoS Routing further enhances performance by integrating real-time energy levels into routing decisions, preventing node depletion and prolonging network lifetime [86]. These protocols, along with other QoS-driven approaches [90], contribute to maintaining performance stability in resource-constrained WSNs.

f: SECURITY-ORIENTED

Security-oriented routing protocols integrate cryptographic and intrusion detection mechanisms to mitigate attacks such as eavesdropping, sinkhole attacks, and data tampering. Security-based routing protocols introduce cryptographic mechanisms, increasing computational energy consumption

[91]. The total energy consumption is

$$E_{\text{secure}} = \sum_{k=1}^{H} \left(E_{\text{transmit},k} + E_{\text{receive},k} + E_{\text{crypto},k} \right). \quad (10)$$

Security-oriented routing protocols, which integrate cryptographic mechanisms and intrusion detection strategies, are crucial in mitigating cyber threats, such as eavesdropping, sinkhole attacks, and data tampering. Secure LEACH (S-LEACH) enhances the original LEACH protocol by integrating authentication and encryption mechanisms, preventing malicious nodes from being selected as CHs and ensuring data integrity [92]. Moreover, Intrusion-Tolerant Routing in WSNs (INSENS) and Intrusion-Tolerant Secure Routing Protocol (ITSRP) have been proposed to strengthen network resilience. INSENS constructs a secure multipath routing framework by employing encryption and redundancy, effectively mitigating attacks such as sinkholes and selective forwarding [93]. ITSRP further enhances security by incorporating authenticated key exchange mechanisms, reducing the risk of unauthorized access while optimizing energy consumption in resource-constrained sensor networks. These protocols represent some examples of security-oriented routing approaches designed to enhance reliability and resilience in WSNs.

The protocols categorized under location-based, multipath, QoS-aware, and security-oriented approaches, along with their routing strategies and key features, are summarized in Table 5.

E. PARAMETER ESTIMATION IN WSNs

Parameter estimation techniques in WSNs are crucial for deriving accurate and meaningful insights from noisy or incomplete data. These methods support various applications, such as localization, environmental monitoring, and fault detection, by ensuring robust and efficient processing in dynamic and resource-constrained environments. This subsection explores four key methodologies: Maximum Likelihood Estimation (MLE), Bayesian Estimation, Kalman Filtering, and Adaptive Estimation using Least Mean Squares (LMS) methods, delving into their theoretical foundations, implementation in WSNs, and real-world applications [94].

1) MAXIMUM LIKELIHOOD ESTIMATION

MLE is a statistical technique widely used in WSNs for parameter estimation tasks such as source localization, signal strength prediction, and environmental variable inference [95], [96]. MLE identifies the parameter θ that maximizes the likelihood function $L(\theta)$, which is defined as:

$$L(\theta) = P(X|\theta), \quad (11)$$

where $P(X|\theta)$ represents the probability of observing the data X given θ. The optimal parameter estimate $\hat{\theta}$ is obtained by solving:

$$\hat{\theta} = \arg\max_{\theta} L(\theta). \quad (12)$$

In WSNs, MLE is particularly effective for source localization based on noisy time-of-arrival (TOA) or received signal strength (RSS) measurements. For example, the source location $\mathbf{p}$ can be estimated by minimizing the squared difference between observed and predicted measurements

$$\hat{\mathbf{p}} = \arg\min_{\mathbf{p}} \sum_{i=1}^{N} (d_i - \|\mathbf{p} - \mathbf{p}_i\|)^2, \quad (13)$$

where d_i is the observed distance from the i-th node, $\mathbf{p}_i$ is the position of the i-th node, and $\|\cdot\|$ denotes the Euclidean distance [97], [98].

While MLE provides statistically efficient estimates, its computational complexity scales with the network size and the dimensionality of the parameter space, posing challenges for real-time applications in resource-constrained environments [99].

2) BAYESIAN ESTIMATION

Bayesian Estimation extends MLE by incorporating prior knowledge about the parameter being estimated, making it particularly useful in dynamic and uncertain environments. The posterior distribution $P(\theta|X)$ is derived using Bayes' theorem

$$P(\theta|X) = \frac{P(X|\theta)P(\theta)}{P(X)}, \quad (14)$$

where $P(\theta)$ is the prior distribution representing prior knowledge, and $P(X)$ normalizes the posterior [100], [101].

In WSNs, Bayesian Estimation is employed in tasks such as environmental monitoring and target tracking. For instance, in estimating the temperature of a region, the prior distribution $P(\theta)$ might reflect seasonal patterns, while $P(X|\theta)$ models the likelihood of real-time sensor measurements. The posterior $P(\theta|X)$ combines these sources of information to provide a refined estimate of the temperature [102], [103].

This method is particularly advantageous for sequential estimation in dynamic systems. For example, in a recursive Bayesian framework, the posterior distribution at time $k-1$ becomes the prior for time k, allowing the system to adapt to changing conditions in real-time. However, Bayesian methods are computationally demanding, especially when the parameter space is large, requiring techniques such as Markov Chain Monte Carlo (MCMC) to approximate the posterior distribution [104], [105], [106].

3) KALMAN FILTERING

Kalman Filtering is a recursive algorithm designed for estimating the state of a linear dynamic system from noisy observations. A state transition model represents the system dynamics

$$\mathbf{x}_k = \mathbf{A}_k \mathbf{x}_{k-1} + \mathbf{w}_k, \quad (15)$$

and an observation model:

$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k, \quad (16)$$

TABLE 5. Classification of routing protocols in WSNs (continued).

Category	Protocol	Routing Approach	Key Features
Location-based	GPSR	Greedy forwarding with perimeter routing	Uses geographical coordinates to minimize hop count and apply perimeter routing when needed
	GEAR	Energy-aware geographic routing	Considers both residual energy and geographic distance for load balancing
	GERAF	Geographic random forwarding	Selects forwarding nodes based on residual energy and location probability
	TBF	Trajectory-based forwarding	Optimizes routing based on predefined trajectories for mobile sink applications
	BVGF	Beacon vector-based forwarding	Improves routing efficiency in dynamic WSN environments using beacon-assisted navigation
	GOAFR	Hybrid geographic routing	Guaranteed delivery adaptive routing local position-based decisions
Multipath	MPDD	Multipath Directed Diffusion	Maintains multiple paths to increase reliability and prevent node failures
	ReInForM	Reliable multi-path routing	Prioritizes packets based on criticality and ensures multiple redundant paths
	MREC	Multipath routing with erasure coding	Improves reliability by encoding redundant data for loss recovery
QoS-aware	SPEED	Real-time delay-aware routing	Uses delay estimation to ensure end-to-end real-time data delivery
	SAR	Priority-based multipath routing	Selects paths considering energy, priority levels, and delay constraints
	EEQR	Energy-efficient QoS routing	Balances energy efficiency and latency using adaptive routing selection
	MCFA	Cost-aware minimum forwarding	Assigns cost metrics based on energy and delay to optimize forwarding paths
	Energy-aware QoS Routing	Real-time energy-based routing	Incorporates real-time energy levels to prevent node depletion
Security	S-LEACH	Secure clustering-based routing	Integrates authentication and encryption to prevent malicious node selection
	INSENS	Intrusion-tolerant secure routing	Uses multipath encryption to ensure secure data transmission
	ITSRP	Intrusion-tolerant secure routing	Employs key exchange mechanisms to enhance security and energy efficiency

where $\mathbf{x}_k$ is the state vector, $\mathbf{A}_k$ is the state transition matrix, $\mathbf{z}_k$ is the observation vector, $\mathbf{H}_k$ is the observation matrix, and $\mathbf{w}_k$, $\mathbf{v}_k$ represent process and observation noise, respectively [107], [108].

Kalman Filtering operates in two stages: prediction and update. The filter estimates the current state using the state transition model in the prediction stage. In the update stage, the estimate is refined by incorporating the observed data. The filter aims to minimize the mean squared error of the state estimate, which makes it highly effective for real-time applications [109], [110].

In WSNs, Kalman Filtering is extensively used for target tracking and environmental monitoring. For instance, tracking a moving object requires estimating its position and velocity over time, fusing noisy observations from multiple nodes. The low computational complexity of the Kalman Filter, combined with its ability to process data in real time, makes it suitable for resource-constrained WSNs [111], [112], [113].

4) ADAPTIVE ESTIMATION TECHNIQUES WITH LEAST MEAN SQUARES

Adaptive estimation techniques based on the LMS algorithm are fundamental for WSNs, effectively addressing dynamic and noisy conditions. These techniques iteratively adjust parameters to minimize error, enabling networks to adapt to environmental changes and system dynamics. The simplicity, computational efficiency, and real-time suitability of the LMS algorithm make it widely applicable in WSNs [114], [115].

The LMS algorithm minimizes the error between the desired signal $d(n)$ and the estimated signal $y(n)$. The error is defined as:

$$e(n) = d(n) - y(n), \tag{17}$$

where $y(n)$ is calculated as:

$$y(n) = \mathbf{w}^T(n)\mathbf{x}(n). \tag{18}$$

Here, $\mathbf{w}(n)$ is the weight vector at iteration n, and $\mathbf{x}(n)$ is the input vector. The weight vector is updated iteratively using the formula:

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \mu e(n)\mathbf{x}(n), \tag{19}$$

where μ is the step size that controls convergence speed and stability [116].

In distributed WSNs, LMS-based techniques are extended to decentralized frameworks, such as Adapt-Then-Combine (ATC) and Combine-Then-Adapt (CTA). The ATC method first adapts the local estimate using node-specific data, followed by combining it with neighbors' estimates. Conversely, CTA prioritizes consensus by combining estimates before local adjustment. These methods enhance performance in distributed environments, balancing local adaptation and network-wide collaboration [117], [118], [119].

Iterative LMS techniques further refine the algorithm by performing multiple weight updates in a single time step. This approach improves accuracy in high-noise or

rapidly changing scenarios but increases computational requirements [120].

Applications of LMS-based techniques include dynamic localization, where nodes estimate mobile targets' positions using noisy data, and environmental monitoring, where parameters such as temperature and humidity are accurately estimated under fluctuating conditions. Fault detection and isolation benefit from LMS methods by identifying anomalies and isolating faulty nodes. In distributed parameter estimation, LMS enables efficient collaboration among nodes to compute global metrics while maintaining energy efficiency [121], [122], [123].

The convergence of LMS algorithms depends on the step size μ, which must satisfy:

$$0 < \mu < \frac{2}{\lambda_{\max}}, \tag{20}$$

where $\lambda_{\max}$ is the largest eigenvalue of the input covariance matrix $\mathbf{R}$. A smaller step size ensures stability but slows convergence, while a larger step size accelerates convergence at the risk of instability [124], [125].

Despite their advantages, LMS-based methods face challenges, such as the need for efficient inter-node communication and parameter optimization, especially in large-scale and resource-constrained WSNs. Nonetheless, their adaptability and scalability make LMS methods integral to modern WSN parameter estimation techniques [126], [127].

III. APPLICATIONS OF WIRELESS SENSOR NETWORKS

WSNs have emerged as transformative technologies across a broad spectrum of domains, leveraging distributed sensing, real-time data processing, and adaptive communication to address complex challenges. Their applications span diverse fields such as environmental monitoring, healthcare, industrial automation, smart cities, precision agriculture, disaster management and emergency response, energy management and renewable energy systems, transportation and logistics, aquatic and underwater monitoring, space exploration and planetary research, UAVs, and military defence. This section presents a detailed examination of their applications, emphasizing scientific advancements and domain-specific methodologies.

A. ENVIRONMENTAL MONITORING

WSNs are fundamental to environmental monitoring due to their capability to collect high-resolution spatiotemporal data from remote and hazardous areas. They are employed for applications like microclimate monitoring, pollutant detection, and disaster prediction [128].

For instance, microclimate monitoring utilizes temperature, humidity, and solar radiation sensors to model localized weather patterns. Predictive algorithms such as recurrent neural networks (RNNs) analyze the temporal dynamics of these parameters, enhancing forecasting accuracy. Pollution monitoring systems deploy gas sensors to measure concentrations of NO_2, CO_2, and particulates, refining their outputs using Bayesian estimation to address sensor inaccuracies caused by environmental drift [129], [130].

In disaster prediction, WSNs play a pivotal role in early warning systems. For example, in landslide detection, pressure and soil moisture sensors are integrated into a distributed network. Data fusion techniques like distributed Kalman filtering combine sensor outputs to predict slope instability. Similarly, flood monitoring networks use ultrasonic and pressure sensors to measure water levels, employing adaptive routing protocols to maintain connectivity even during infrastructural damage [131], [132], [133].

B. HEALTHCARE AND BIOMEDICAL APPLICATIONS

WSNs have revolutionized patient care and medical diagnostics in healthcare through real-time monitoring and predictive analysis. Body Sensor Networks (BSNs), a specialized WSN application, enable continuous tracking of physiological parameters such as electrocardiograph (ECG), SpO_2, and glucose levels [134].

Advanced applications include early detection of critical health events using AI-powered analysis. For instance, deep learning models such as convolutional neural networks (CNNs) and long short-term memory (LSTM) networks process real-time data to identify arrhythmias or hypoglycemia. Wearable devices in BSNs are often equipped with energy harvesting technologies, such as thermoelectric generators (TEGs) that utilize body heat, ensuring long-term operation [135], [136], [137].

Integration with telemedicine platforms allows sensor data to be transmitted securely to healthcare providers using cryptographic protocols such as the Advanced Encryption Standard (AES). This facilitates remote diagnosis and proactive interventions, particularly in chronic disease management and post-operative care [138], [139].

C. INDUSTRIAL AUTOMATION

In industrial automation, WSNs enhance efficiency and reliability by enabling predictive maintenance, real-time monitoring, and automated control of processes. Sensor nodes monitor parameters such as vibration, temperature, and pressure in industrial equipment [140].

An advanced example is the use of Fast Fourier transform (FFT) and wavelet transform-based analysis in vibration monitoring to detect early signs of mechanical failures. These techniques decompose vibration signals into frequency components, identifying patterns indicative of wear or misalignment. Hierarchical network architectures are commonly employed, where edge nodes perform initial data processing and central nodes handle predictive modeling using ML algorithms [141], [142], [143].

Protocols like WirelessHART and ISA100.11a provide reliable communication in noisy industrial environments, ensuring low latency and robustness. Time-sensitive networks (TSNs) integrated with WSNs enable precise synchronization, which is critical in robotic assembly lines where sub-millisecond accuracy is required [144], [145].

D. SMART CITIES

Smart city applications of WSNs optimize urban services and infrastructure, addressing challenges such as traffic congestion, energy consumption, and public safety. Traffic management systems deploy WSNs to monitor vehicle density, speed, and flow, using this data for dynamic traffic light control and real-time navigation updates [146], [147].

Predictive models based on support vector regression (SVR) or gradient-boosted decision trees analyze traffic patterns, enabling the system to adjust to anticipated congestion preemptively. In energy-efficient street lighting systems, WSN nodes equipped with motion and light sensors dynamically adjust illumination levels, reducing energy consumption [148], [149], [150].

Public safety applications include seismic monitoring networks and structural health assessment systems. Accelerometer-based WSNs detect vibrations during earthquakes, employing distributed event detection algorithms to trigger early warnings. Such systems are integrated with centralized dashboards, enabling rapid decision-making during emergencies [151], [152].

E. PRECISION AGRICULTURE

WSNs have transformed agriculture by providing real-time insights into soil, crop, and environmental conditions. Applications range from irrigation optimization to automated pest control and yield prediction [153].

Soil moisture monitoring networks utilize capacitive and resistive sensors deployed at varying depths. These data are processed using regression models and weather forecast integrations to create adaptive irrigation schedules. Pest monitoring systems combine pheromone traps with image sensors, using CNNs to classify pests and estimate infestation levels. This data informs targeted pesticide application, reducing costs and environmental impact [154], [155], [156].

Crop yield prediction employs multispectral imaging sensors and ML models such as random forests and gradient boosting. These systems analyze vegetation indices like NDVI (Normalized Difference Vegetation Index), enabling farmers to make data-driven decisions to maximize productivity [157], [158].

F. MILITARY AND DEFENSE APPLICATIONS

In defense, WSNs support battlefield surveillance, intrusion detection, and logistics management. Deployed in hostile environments, these networks provide real-time situational awareness through acoustic and seismic sensors. Signal processing techniques such as short-time Fourier transform (STFT) analyze acoustic signatures to differentiate between personnel, vehicles, and drones [159], [160], [161].

Logistics applications integrate WSNs with Radio Frequency Identification (RFID) tags to monitor the condition and location of supplies. Hybrid networks combining ground sensors and UAVs ensure extended coverage and resilience in dynamic combat scenarios. Secure communication protocols such as elliptic curve cryptography (ECC) protect sensitive data, ensuring operational integrity [162], [163].

G. DISASTER MANAGEMENT AND EMERGENCY RESPONSE

WSNs are critical in disaster management, offering early warnings and situational awareness. Landslide monitoring networks employ distributed sensor nodes to measure soil pressure and moisture, combining data using predictive algorithms to estimate risk levels. Similarly, flood monitoring systems use ultrasonic sensors to track water levels, with adaptive routing ensuring data transmission under adverse conditions [164], [165], [166].

Emergency response applications include victim localization, where WSNs equipped with infrared and acoustic sensors identify survivors in collapsed structures. These networks integrate with centralized control systems, enabling efficient resource allocation and response coordination [167], [168].

H. ENERGY MANAGEMENT AND RENEWABLE ENERGY SYSTEMS

WSNs are integral to energy systems, monitoring parameters such as solar irradiance, wind speed, and energy output in renewable energy setups. Photovoltaic panel monitoring networks detect inefficiencies due to shading or dust accumulation, triggering automated maintenance alerts [169], [170].

In wind farms, vibration sensors track turbine performance, with predictive maintenance algorithms preemptively addressing potential failures. Smart grids employ WSNs for real-time load balancing and fault detection. ML models predict energy demand patterns, optimizing the integration of renewable energy sources into the grid [171], [172], [173].

I. TRANSPORTATION AND LOGISTICS

WSNs optimize transportation infrastructure and logistics by providing real-time monitoring and predictive insights. Railway track monitoring networks use vibration and displacement sensors to detect structural abnormalities, ensuring operational safety [174], [175].

In logistics, sensor-equipped cargo containers monitor environmental conditions like temperature and humidity, safeguarding perishable goods. These systems integrate with Global Positioning System (GPS) platforms, enabling real-time tracking and route optimization. Predictive analytics enhances delivery efficiency, reducing delays and resource wastage [176], [177], [178].

J. AQUATIC AND UNDERWATER MONITORING

Aquatic WSNs monitor water quality, oceanographic parameters, and ecosystem health. Applications include detecting oil spills, monitoring dissolved oxygen levels, and assessing salinity gradients. In aquaculture, these networks optimize feeding schedules and water conditions, enhancing yield [179], [180].

Underwater WSNs rely on acoustic communication to overcome the attenuation of radio waves in water. Advanced power management and adaptive routing protocols address the high latency and energy constraints inherent in underwater environments [181], [182].

K. SPACE EXPLORATION AND PLANETARY RESEARCH

WSNs are being increasingly considered for autonomous exploration of extraterrestrial environments. On planetary surfaces, distributed sensor nodes measure radiation levels, surface composition, and temperature gradients. NASA and other agencies employ WSNs in missions to enable autonomous data collection over vast areas [183], [184], [185]. Spacecraft systems also rely on WSNs for structural health monitoring, ensuring operational safety during long-duration missions. Fault-tolerant communication protocols maintain data reliability, even in the harsh space conditions [186], [187].

L. UNMANNED AERIAL VEHICLES

UAVs have emerged as a crucial application of WSNs, enabling real-time data collection, environmental monitoring, and disaster response. By integrating WSNs with UAVs, autonomous or remotely operated aerial platforms can gather sensor data over vast and inaccessible areas, enhancing situational awareness and decision-making. These systems are particularly valuable for applications such as precision agriculture, where UAVs equipped with sensors monitor crop health, detect anomalies, and optimize resource distribution. Similarly, in search and rescue missions, UAVs can quickly survey disaster-stricken regions, identify survivors, and relay critical information to ground teams, significantly reducing response times [188], [189], [190].

Beyond emergency and environmental applications, UAV-based WSNs play a vital role in military surveillance, infrastructure inspection, and smart city development. In defense operations, UAVs equipped with sensor networks enhance battlefield intelligence by providing real-time reconnaissance and threat detection. For infrastructure maintenance, such as monitoring bridges, power lines, and pipelines, UAVs offer a cost-effective solution for identifying structural issues without requiring manual inspections. Additionally, in urban settings, UAVs contribute to smart city initiatives by monitoring air pollution, traffic congestion, and public safety, leveraging WSNs to create data-driven, efficient urban management systems [191], [192], [193], [194].

Table 6 provides a comparative summary of WSN applications, highlighting key features, technologies, use cases, and references across various domains. The table categorizes applications in areas such as environmental monitoring, healthcare, industrial automation, smart cities, precision agriculture, military, disaster management, energy management, transportation, underwater monitoring, space exploration, and UAVs. Each application is associated with specific technologies and methodologies, such as Bayesian estimation and Kalman filtering for environmental monitoring, AI-driven health analytics for biomedical applications, and IoT-integrated predictive maintenance in industrial settings. The table also emphasizes innovations like AI-based pest monitoring in agriculture, blockchain-secured communications in military applications, and quantum sensors for space exploration, showcasing the advancements driving WSN adoption in real-world scenarios.

The analysis of Table 6 is complemented by Table 7, focusing on real-world deployments. More specifically, Table 7 presents domain-specific case studies that illustrate how WSNs have been implemented in practice. These include named projects and deployments such as the Internet of Forests (IoF) initiative in the Amazon, smart grid monitoring in India, and the European Space Agency's (ESA's) PANGAEA (Pan and Gaea) training program for space exploration. Each example grounds the application domain in a tangible setting, thereby offering concrete evidence of WSN adoption and impact. Together, these tables provide a holistic perspective, where the former contextualizes technological trends and innovations, whereas the latter demonstrates their translation into practical, operational systems.

IV. INNOVATIONS IN WSN TECHNOLOGIES

Breakthroughs in energy systems, communication protocols, data processing frameworks, and integration with emerging paradigms such as the IoT and AI have propelled WSNs' evolution. These innovations address core challenges like energy constraints, dynamic environments, security vulnerabilities, and the need for real-time processing. This section explores how these advancements are enabling WSNs to operate more efficiently, intelligently, and securely, driving their application in increasingly complex scenarios.

A. ENERGY HARVESTING AND AUTONOMOUS POWER MANAGEMENT

Energy availability is a defining constraint for WSNs. Innovations in energy harvesting technologies have shifted the paradigm from battery dependency to autonomous power systems. Solar cells remain the most mature technology, with advancements in Maximum Power Point Tracking (MPPT) algorithms improving efficiency under variable light conditions. For industrial environments, piezoelectric systems harvest vibrational energy from machinery, while Thermoelectric Generators (TEGs) utilize temperature gradients in high-heat regions such as manufacturing plants or engines [207], [208], [209].

Complementing these harvesting methods, ultra-low-power design principles in hardware and software ensure minimal energy wastage. Dynamic power scaling, duty-cycling mechanisms, and wake-up radios are now standard for managing operational energy. For instance, wake-up radios maintain nodes in a low-power state and activate them only upon detecting relevant communication signals, reducing idle listening [210], [211].

TABLE 6. Comparative summary of WSN applications and technologies.

Application Domain	References	Key Features	Technologies–Methods	Example Use Cases
Environmental Monitoring	[128] [129] [130] [131] [132] [133]	High-resolution spatiotemporal data collection; remote and hazardous areas monitoring	Microclimate monitoring, Bayesian estimation, Kalman filtering, predictive algorithms (e.g., RNNs)	Landslide detection, pollutant monitoring, flood prediction
Healthcare & Biomedical	[134] [135] [136] [137] [138] [139]	Real-time physiological monitoring; predictive health analytics	BSNs, AI-powered models (CNNs, LSTMs), cryptographic protocols (e.g., AES)	Chronic disease management, remote post-operative care, hypoglycemia detection
Industrial Automation	[140] [141] [142] [143] [144] [145]	Predictive maintenance, real-time monitoring, and automated control	FFT, wavelet transforms, hierarchical architectures, WirelessHART, ISA100.11a	Vibration monitoring, robotic assembly line synchronization, energy-efficient process automation
Smart Cities	[146] [147] [148] [149] [150] [151] [152]	Urban optimization; real-time traffic and energy management	SVR, gradient-boosted decision trees, motion/light sensors, distributed systems	Dynamic traffic light control, energy-efficient street lighting, seismic monitoring
Precision Agriculture	[153] [154] [155] [156] [157] [158]	Data-driven insights for optimizing irrigation, pest control, and crop yields	Capacitive sensors, CNNs, NDVI analysis, random forests, gradient boosting	Adaptive irrigation schedules, pest classification, yield prediction
Military & Defense	[159] [160] [161] [162] [163]	Real-time situational awareness; secure and resilient communication	STFT, UAV-integrated hybrid networks, ECC encryption	Battlefield surveillance, logistics management, intrusion detection
Disaster Management	[164] [165] [166] [167] [168]	Early warning systems; situational awareness in emergencies	Kalman filtering, adaptive routing protocols, infrared/acoustic sensors	Flood monitoring, landslide prediction, victim localization in collapsed structures
Energy Management	[169] [170] [171] [172] [173]	Renewable energy monitoring and optimization	Predictive maintenance algorithms, vibration sensors, WSN-integrated smart grids	Wind turbine monitoring, solar panel efficiency optimization, real-time load balancing
Transportation & Logistics	[174] [175] [176] [177] [178]	Infrastructure safety and cargo condition monitoring	Vibration–displacement sensors, GPS-integrated systems, predictive analytics	Railway track monitoring, perishable goods monitoring, route optimization
Aquatic & Underwater	[179] [180] [181] [182]	Ecosystem health monitoring and high-latency communication solutions	Acoustic communication, power management, adaptive routing protocols	Oil spill detection, water quality assessment, aquaculture optimization
Space Exploration	[183] [184] [185] [186] [187]	Autonomous planetary exploration; fault-tolerant communication	Radiation-hardened hardware, interplanetary protocols	Surface composition analysis, spacecraft structural health monitoring
UAVs	[188] [189] [190] [191] [192] [193] [194]	Real-time data collection, autonomous monitoring, and adaptive communication	Wireless sensor integration, AI-based data analytics, energy-efficient routing, edge computing for onboard processing	Disaster response, smart surveillance, environmental monitoring, precision agriculture, military reconnaissance

Recent innovations extend into hybrid energy systems that combine multiple harvesting sources, providing resilience in environments with fluctuating energy availability. Predictive models optimize the allocation of harvested energy, enabling nodes to schedule tasks based on energy budgets and environmental conditions [212], [213].

Despite these innovations, current energy harvesting systems remain constrained by environmental variability, low power density, and limited storage efficiency. These factors impact the stability and scalability of WSN deployments, particularly in harsh or unpredictable settings. Furthermore, while hybrid and predictive models improve reliability, their real-world implementation still demands lightweight, energy-aware computation. These limitations are further explored in Section V, where dynamic optimization strategies and future research directions are discussed.

B. ADAPTIVE COMMUNICATION PROTOCOLS

Communication in WSNs must balance energy consumption, latency, and reliability, particularly in challenging environments like underwater, industrial, or dense urban areas. Adaptive communication protocols increasingly employ dynamic channel selection, power control, and modulation schemes to optimize performance [214], [215].

Time-Slotted Channel Hopping, part of IEEE 802.15.4e, has revolutionized communication in industrial WSNs by introducing time synchronization and channel hopping. This reduces interference and packet collisions, ensuring reliable data delivery in noisy environments. In underwater sensor networks, hybrid acoustic-optical systems combine the range of acoustic communication with the high bandwidth of optical methods, dynamically switching based on environmental conditions [216], [217], [218].

TABLE 7. Recent real-world case studies of WSN deployments across application domains.

Application Domain	Specific Use Case	Real-world Case Study
Environmental Monitoring	Illegal logging detection and forest conservation	Deployment of the IoF by ONE AMAZON, Veea, and AECOM [195]
Healthcare and Biomedical Applications	Smart hospital sensor networks	WSN-based monitoring of patient vitals and environment in hospitals [196]
Industrial Automation	Pipeline monitoring and vandalism detection	WSN deployment in Nigeria's oil and gas industry [197]
Smart Cities	Urban environmental monitoring	WSN use in Cary, North Carolina, for smart water and flood alert systems [198]
Precision Agriculture	Soil moisture and nutrient monitoring	WSN-based nutrient sensing in tomato fields [199]
Military and Defense	Tactical edge computing and mission-aware sensing	NATO's integrated Modular Unmanned Ground System (iMUGS) project with WSNs for Unmanned Ground Vehicles (UGVs) [200]
Disaster Management and Emergency Response	Flood detection and victim localization	WSNs for flood emergency response coordination [201]
Energy Management and Renewable Energy Systems	Smart grid monitoring and energy optimization	Real-time smart grid monitoring deployment in India [202]
Transportation and Logistics	Port operations and cargo tracking	Altai Super WiFi™ network at Angamos Port, Chile [203]
Aquatic and Underwater Monitoring	Marine environment monitoring and subsea infrastructure	Battery-free subsea IoT for marine sensing [204]
Space Exploration and Planetary Research	Lunar exploration and sample collection	ESA's PANGAEA WSN training program [205]
UAVs	Sensor positioning and data collection in remote zones	UAV-assisted WSNs for extended sensing coverage [206]

Protocols like 6LoWPAN and RPL have enabled seamless integration of WSNs into IoT systems by supporting IPv6, ensuring scalability across large, heterogeneous networks. These protocols incorporate topology optimization to maintain connectivity in dynamic environments, improving network stability and efficiency [219], [220].

C. EDGE INTELLIGENCE AND DISTRIBUTED DATA PROCESSING

The integration of edge intelligence in WSNs has transformed their ability to operate autonomously and adaptively in real-time. By embedding computation capabilities within sensor nodes, data can now be processed locally, reducing the need for centralized processing and thereby minimizing latency and bandwidth usage [221], [222].

Tiny ML (TinyML) frameworks have successfully deployed AI models on resource-constrained sensor nodes. For instance, CNNs trained for image classification are optimized for deployment on microcontrollers, enabling applications like real-time intrusion detection and anomaly identification directly at the node level. FL has further enhanced distributed intelligence by allowing collaborative model training across nodes without sharing raw data, addressing both privacy concerns and bandwidth limitations [223], [224], [225].

Fog computing extends these capabilities by acting as an intermediary between sensor nodes and the cloud. This hierarchical processing framework dynamically allocates tasks to nodes, gateways, and edge servers, optimizing resource usage and ensuring resilience to failures in any single layer [226], [227].

D. SECURITY AND TRUST MECHANISMS

Security is critical as WSNs are increasingly used in sensitive applications such as healthcare, military, and critical infrastructure. Lightweight cryptography has emerged as a cornerstone, with algorithms like ECC offering robust encryption without taxing computational resources [228], [229], [230].

Blockchain technologies are being explored to establish trust in decentralized WSNs. For example, smart contracts can automate access control policies, ensuring that only authorized nodes interact with sensitive data. Blockchain-based authentication protocols enhance node identity management, preventing malicious actors from infiltrating the network [231], [232], [233].

IDS powered by ML analyzes network traffic patterns to identify anomalies indicative of attacks, such as denial of service (DoS) or sinkhole attacks. These systems leverage features like packet flow, energy usage, and delay metrics to build accurate threat detection models [234], [235].

E. INTEGRATION WITH IoT, AI, AND DIGITAL TWINS

The convergence of WSNs with IoT and AI has enabled autonomous systems that are both scalable and intelligent. IoT frameworks like AWS IoT and Azure IoT Hub aggregate data from WSNs for cloud-based analytics, providing centralized dashboards and enabling large-scale monitoring [236], [237].

AI enhances the operational intelligence of WSNs. For instance, deep RL optimizes resource allocation by learning dynamic policies for task scheduling and energy management. Predictive maintenance systems use AI to analyze sensor data, identifying early signs of equipment failure and triggering preemptive actions [238], [239].

Digital Twin technology integrates WSN data into virtual replicas of physical systems, enabling advanced simulations for predictive maintenance, scenario testing, and operational optimization. These virtual models incorporate real-time updates from WSNs, allowing for proactive decision-making in industries such as manufacturing, energy, and urban planning [240], [241], [242].

F. SOFTWARE-DEFINED WSNs

A promising research direction involves Software-Defined WSNs (SDWSNs), which separate the control and data planes to enable centralized policy enforcement, dynamic reconfiguration, and cross-layer optimization [243]. In this architecture, a logically centralized controller governs network behavior through programmable interfaces, improving agility, scalability, and manageability.

Frameworks like SD-WISE exemplify how network function virtualization and trusted hardware can extend SDN concepts to resource-constrained sensor nodes [244]. However, real-world deployment faces persistent challenges: the overhead of in-band control signaling, scalability limits of southbound protocols, and the lack of lightweight, fault-tolerant controllers tailored to WSN dynamics [245], [246].

Future directions should emphasize the development of energy-efficient southbound protocols, modular and resilient controller architectures, and integrated support for in-network processing and mobility. These advances will be critical to deploying SDWSNs in dynamic, large-scale, and delay-sensitive applications.

G. ADVANCES IN HARDWARE DESIGN AND SENSING TECHNOLOGIES

Innovations in hardware have significantly expanded the capabilities of WSNs. Ultra-low-power microcontrollers, such as those in the ARM Cortex-M series, now support complex computations while minimizing energy consumption. Transceivers based on LoRa technology enable long-range, low-bit-rate communication ideal for wide-area deployments, while IEEE 802.15.4-compliant radios support low-power communication suitable for dense sensor networks. Higher-layer protocols such as Zigbee and 6LoWPAN, built on these physical layers, facilitate interoperability, self-organization, and energy-aware routing in large-scale applications such as smart agriculture and urban sensing [247], [248], [249], [250].

Multi-modal sensors now integrate capabilities such as imaging, gas detection, and environmental sensing in a single compact device. These sensors generate rich datasets, supporting applications like real-time environmental monitoring and advanced robotics [251], [252].

Emerging technologies such as nanosensors enable molecular-level detection and revolutionize healthcare and environmental monitoring by allowing the detection of pathogens or pollutants at extremely low concentrations. Flexible and stretchable electronics provide solutions for wearable devices, enabling continuous health monitoring with minimal discomfort [253], [254], [255].

H. SMART SENSORS AND INTELLIGENT DATA PROCESSING

Integrating smart sensors represents a significant innovation in WSNs, enabling real-time data processing and reducing dependency on centralized computation. Unlike traditional sensors that passively collect and transmit data, smart sensors incorporate embedded processors that filter, analyze, and preprocess information locally. This advancement minimizes redundant transmissions, lowers bandwidth usage, and enhances the responsiveness of WSNs. By dynamically adjusting their sensing and transmission behavior based on environmental conditions, these sensors optimize energy consumption, contributing to prolonged network lifetime and more sustainable deployments [256], [257], [258].

A key innovation in smart sensors is their ability to support edge intelligence within WSNs. These sensors can identify patterns, detect anomalies, and prioritize critical data at the source by integrating lightweight ML models and adaptive signal processing techniques. This reduces network congestion and computational overhead at cloud servers while enabling real-time decision-making in latency-sensitive applications. Advanced wireless communication protocols, including IEEE 802.15.4, Zigbee, and LoRaWAN, allow smart sensors to implement self-organizing and adaptive networking strategies. These mechanisms dynamically regulate transmission power and frequency, ensuring efficient spectrum utilization and resilience in complex WSN environments [259], [260], [261].

Further innovations in smart sensors include the development of ultra-low-power hardware architectures and neuromorphic computing for on-sensor intelligence. These advancements facilitate energy-efficient, event-driven sensing, allowing WSNs to operate autonomously with minimal external intervention. Secure data transmission is also a growing focus, with the adoption of blockchain-based authentication and lightweight cryptographic techniques to enhance network trust and reliability. As research progresses, smart sensors will continue to play a pivotal role in the evolution of WSNs, driving improvements in scalability, efficiency, and adaptability to dynamic operational conditions [262], [263], [264], [265].

Beyond industrial and environmental applications, smart sensors play a crucial role in smart home and smart building automation. WSNs enhance energy efficiency, security, and user comfort by enabling real-time monitoring and adaptive control of indoor environments. Sensor nodes deployed in smart buildings track temperature, humidity, air quality, and occupancy, optimizing HVAC (Heating, Ventilation, and Air Conditioning) systems for reduced energy consumption. Furthermore, smart homes integrate motion sensors, door/window sensors, and smart meters to facilitate energy-aware automation, intrusion detection, and fire safety. With

advancements in edge computing and AI-driven analytics, smart sensors in intelligent buildings are evolving toward self-learning systems capable of predictive maintenance and real-time anomaly detection, improving sustainability and security [266].

Table 8 provides a comparative analysis of innovations in WSN technologies, outlining key advancements, methodologies, and their impact on various applications. It categorizes innovations such as energy harvesting, adaptive communication protocols, edge intelligence, security mechanisms, IoT integration, software-defined networking, and advanced sensing technologies. The table highlights how these advancements enhance network efficiency, scalability, security, and real-time processing, enabling improved performance in domains like smart cities, healthcare, industrial automation, and environmental monitoring. By summarizing these technological developments, the table offers insights into emerging trends shaping next-generation WSN deployments.

V. CHALLENGES IN WSN APPLICATIONS

WSNs face numerous challenges that constrain their performance, scalability, and reliability. These issues arise from the inherent resource limitations of sensor nodes, dynamic deployment environments, and the increasing complexity of modern applications. Addressing these challenges requires a scientific understanding of their root causes and the development of targeted solutions. This section provides a detailed analysis of the technical and operational barriers to WSN deployment and functionality.

Many of the challenges discussed in this section have been recognized since the early development of WSNs. However, they remain critical today, hindering system scalability, energy autonomy, and long-term reliability in real-world deployments. Their persistence, despite continuous innovation, highlights the need for renewed focus and interdisciplinary solutions tailored to emerging application contexts.

A. ENERGY LIMITATIONS AND DYNAMIC POWER OPTIMIZATION

Energy availability remains the most critical challenge in WSNs, directly affecting the longevity and functionality of deployments. Despite advancements in energy harvesting systems, their performance depends on environmental conditions, such as light availability for solar cells or temperature gradients for TEGs. This dependency introduces variability in energy supply, complicating power management in dynamic environments [267], [268].

Dynamic power optimization techniques attempt to address these limitations by adapting node operations based on real-time energy levels and task requirements. For example, stochastic models predict energy availability in hybrid harvesting systems, allowing nodes to adjust their duty cycles and communication schedules dynamically. However, implementing these techniques requires computational resources, which further strain node energy reserves. Future research must focus on lightweight algorithms that balance computational overhead with power-saving benefits [269], [270], [271].

B. NETWORK SCALABILITY AND CONGESTION CONTROL

Scaling WSNs to larger deployments introduces significant challenges in maintaining efficient communication and data processing. As the number of nodes increases, the network experiences higher congestion levels, packet collisions, and delays. Hierarchical clustering and tree-based routing protocols alleviate these issues by organizing nodes into manageable sub-networks, but they often require centralized control, which creates bottlenecks near the sink [272], [273].

Advanced approaches, such as adaptive clustering, utilize ML models to optimize cluster formation dynamically based on traffic patterns and energy levels. Additionally, cross-layer optimization techniques align routing, MAC protocols, and transport layers to reduce congestion holistically. These methods, while promising, demand computational and memory resources that many sensor nodes lack [274], [275], [276].

C. ENVIRONMENTAL INTERFERENCE AND DEPLOYMENT RESILIENCE

WSNs often operate in environments with fluctuating conditions, such as industrial sites with electromagnetic interference or underwater systems with high signal attenuation. Environmental factors degrade communication links and sensor accuracy, necessitating robust error correction and reconfiguration mechanisms [277], [278].

Deployment resilience relies on redundancy and self-healing algorithms that reroute traffic and redistribute tasks among operational nodes during failures. For instance, network coding techniques improve data recovery by encoding transmitted packets, reducing the impact of packet loss. Furthermore, autonomous deployment strategies using UAVs extend network coverage and address node placement inaccuracies, but their real-time integration with existing WSNs remains an area of active research [279], [280], [281].

D. SECURITY VULNERABILITIES IN RESOURCE-CONSTRAINED NODES

WSNs' decentralized architecture, combined with their reliance on wireless communication, makes them vulnerable to a wide array of attacks, including eavesdropping, jamming, and sinkhole attacks. Conventional security solutions, such as public-key cryptography, are often unsuitable for WSNs due to their high computational and energy costs. Emerging lightweight cryptographic algorithms, such as ChaCha20 and SPECK, offer energy-efficient alternatives for encryption and authentication in resource-constrained WSN nodes [282], [283]. Additionally, blockchain-based distributed trust mechanisms are being explored to secure decentralized WSN deployments [284], [285]. These mechanisms typically rely

TABLE 8. Comparative summary of innovations in WSN technologies.

Innovation Area	References	Key Features	Technologies/Methods	Applications/Impact
Energy Harvesting and Autonomous Power Management	[207] [208] [209] [210] [211] [212] [213]	Shift from battery dependency to self-sustaining systems; use of multiple energy harvesting methods	Solar cells with MPPT algorithms, piezoelectric systems, TEGs, hybrid energy systems	Long-term operation in industrial setups, smart cities, and remote sensing where energy is limited
Adaptive Communication Protocols	[214] [215] [216] [217] [218] [219] [220]	Balance between energy consumption, latency, and reliability in diverse environments	TSCH (IEEE 802.15.4e), hybrid acoustic-optical systems, 6LoWPAN, RPL, dynamic channel selection	Industrial IoT, underwater communication, and urban dense environments requiring reliable data transfer
Edge Intelligence and Distributed Data Processing	[221] [222] [223] [224] [225] [226] [227]	Processing data locally to reduce latency and bandwidth use	TinyML frameworks, FL, fog computing	Real-time anomaly detection, predictive maintenance, and intrusion detection
Security and Trust Mechanisms	[228] [229] [230] [231] [232] [233] [234] [235]	Lightweight encryption and trust models to enhance WSN security	ECC, blockchain-based trust management, ML-powered IDS	Critical infrastructure, healthcare, and military applications requiring secure data transfers
Integration with IoT, AI, and Digital Twins	[236] [237] [238] [239] [240] [241] [242]	Enhanced interoperability with IoT systems and real-time simulation for predictive insights	IoT frameworks (AWS IoT, Azure IoT Hub), DRL, Digital Twin platforms	Advanced predictive systems, industrial operations, and urban planning simulations
Software Defined WSNs	[243] [244] [245] [246]	Logical separation of control and data planes; centralized policy enforcement; dynamic network reconfiguration	Lightweight Southbound protocols, scalable control-plane designs, in-band secure signaling mechanisms tailored for resource-limited nodes	Programmable WSNs for agile IoT applications, dynamic reconfiguration in industrial automation, smart cities, and disaster response
Advances in Hardware and Sensing Technologies	[247] [248] [249] [250] [251] [252] [253] [254] [255]	High-performance, low-power hardware designs and advanced sensor capabilities	ARM Cortex-M series microcontrollers, LoRa/Zigbee transceivers, nanosensors, stretchable electronics	Wearable health monitors, environmental monitoring, and molecular-level detection in healthcare
Smart Sensors and Intelligent Data Processing	[256] [257] [258] [259] [260] [261] [262] [263] [264] [265] [266]	Localized data processing, adaptive sensing, real-time decision-making	Edge AI, neuromorphic computing, blockchain-secured data transmission, adaptive networking protocols (LoRa, Zigbee, IEEE 802.15.4)	Autonomous WSNs, industrial automation, smart healthcare, smart home/building and precision agriculture

on consensus algorithms and smart contracts to authenticate nodes and validate transactions.

However, integrating blockchain technologies with lightweight sensor platforms presents significant challenges. Traditional Proof-of-Work (PoW) mechanisms—widely adopted in public blockchains—are computationally intensive and energy-inefficient, making them fundamentally impractical for WSNs [286]. In response, alternative consensus models such as Proof-of-Stake (PoS) and reputation-based schemes have gained attention due to their reduced computational demands [287]. Yet PoS introduces its own limitations, such as the need for persistent node identity and continuous state tracking, which conflict with the ephemeral, mobile, and intermittently connected characteristics of typical WSN nodes [286]. These trade-offs highlight the pressing need for application-specific consensus protocols that can balance security, energy efficiency, and fault tolerance, tailored to the constraints of real-world WSN deployments [285], [287].

E. APPLICATION-AWARE SECURITY AND PRIVACY CONSIDERATIONS

The application domain strongly influences security and privacy challenges in WSNs. In healthcare, where WSNs support applications such as body sensor networks and remote monitoring, the sensitivity of collected physiological data demands privacy protection beyond secure transmission alone. Unauthorized access or inference attacks on patient data can have profound ethical and legal implications, especially under regulatory frameworks like HIPAA (Health Insurance Portability and Accountability Act) and GDPR (General Data Protection Regulation). Systems must balance responsiveness with privacy-preserving features such as data anonymization, selective data disclosure, and secure user authentication, without exceeding medical sensor nodes' energy and processing limitations [288].

Similarly, in smart homes, behavioral data captured by sensor nodes poses privacy risks such as unauthorized surveillance and profiling if not adequately protected. In industrial automation, WSNs face cyber-physical threats including false data injection, actuator compromise, and availability-targeted attacks, which may disrupt safety-critical systems. These domain-specific risks underscore the importance of implementing security strategies that are both resource-aware and application-context sensitive [289].

F. DATA OVERLOAD AND REAL-TIME ANALYTICS

Modern WSN deployments often produce vast amounts of heterogeneous data, particularly in multi-modal sensing systems. This data deluge overwhelms the limited storage and processing capacities of sensor nodes, leading to bottlenecks in real-time analytics [290], [291].

Edge computing mitigates some of these challenges by enabling data preprocessing at intermediate nodes. For instance, FL models can train predictive algorithms across distributed nodes without centralizing data, reducing

bandwidth requirements and preserving privacy. However, the integration of these models into constrained environments necessitates further optimization of computational efficiency and communication overhead [292], [293], [294].

G. STANDARDIZATION AND INTEROPERABILITY

The proliferation of proprietary protocols and hardware in WSNs complicates the integration of heterogeneous systems, especially in large-scale deployments involving multiple vendors. Standardization efforts, such as the adoption of IPv6 protocols (e.g., 6LoWPAN) and the CoAP (Constrained Application Protocol), aim to improve interoperability but often require retrofitting legacy systems, which is costly and time-intensive [295], [296].

Middleware solutions that abstract device-specific details and provide a unified interface are emerging as potential solutions. However, these frameworks must balance abstraction with performance, ensuring that the underlying system efficiency is not compromised in favor of compatibility [297], [298].

H. LATENCY AND QUALITY OF SERVICE CONSTRAINTS

Applications requiring real-time response, such as autonomous vehicles or industrial automation, demand ultra-low latency and guaranteed QoS. The inherently low-power and bandwidth-constrained nature of WSNs complicates meeting these requirements [299], [300].

For example, delay-sensitive applications in industrial control rely on Time-Sensitive Networking (TSN) extensions integrated with WSN protocols. These systems prioritize critical data flows while employing predictive scheduling algorithms to minimize latency. Despite these advancements, achieving sub-millisecond latency in energy-constrained nodes remains a significant technical hurdle [301], [302], [303].

I. COST-EFFICIENCY AND MAINTENANCE CHALLENGES

The deployment and maintenance of WSNs are resource-intensive, particularly in remote or hazardous locations. Battery replacement, sensor recalibration, and node recovery operations contribute to high operational costs. Autonomous deployment using robotic systems reduces manual effort but introduces additional complexity in coordinating node placement and maintaining connectivity [304], [305].

Predictive maintenance models, trained on historical data, offer a proactive approach to reducing downtime and extending the operational lifespan of nodes. However, implementing these models requires consistent data availability and computational resources, which may not always be feasible in constrained environments [306], [307].

The challenges WSNs face highlight the need for innovative solutions that address energy constraints, scalability, security, and operational resilience. Researchers can overcome these barriers by advancing techniques in dynamic power optimization, adaptive protocols, and real-time analytics, enabling WSNs to meet the demands of modern applications. Addressing these challenges is essential for unlocking the full potential of WSNs in transforming scientific, industrial, and societal domains [308], [309].

Table 9 presents a comparative analysis of challenges in WSNs applications, outlining key issues, proposed solutions, and their impact. It categorizes challenges such as energy limitations, network scalability, environmental interference, security vulnerabilities, data overload, standardization, latency constraints, and cost-efficiency. The table highlights strategies like energy harvesting, adaptive clustering, cryptographic security, and edge computing to address these challenges. By summarizing technological and operational obstacles, the table provides insights into WSN resilience, efficiency, and deployment sustainability across various applications.

VI. FUTURE TRENDS AND RESEARCH DIRECTIONS

Advancements in hardware, communication protocols, and data processing techniques drive WSNs' continuous evolution, as well as their integration with emerging technologies like AI, IoT, and edge computing. This section outlines the future trends and research directions in WSNs, emphasizing innovations that address existing limitations and expand their applications to new frontiers.

A. ULTRA-LOW-POWER AND SELF-SUSTAINING SYSTEMS

Future WSNs will increasingly rely on ultra-low-power designs and self-sustaining energy systems to overcome the constraints of battery-powered nodes. Advancements in nanotechnology are enabling the development of nanoscale energy harvesters that convert ambient sources, such as thermal, vibrational, and electromagnetic energy, into usable power with unprecedented efficiency [310], [311].

Self-sustaining nodes will integrate multi-source energy harvesting with adaptive energy management algorithms that predict environmental energy availability and adjust operational parameters accordingly. For instance, RL-based energy schedulers can optimize duty cycles dynamically, ensuring continuous operation even in environments with fluctuating energy sources. Research into micro-energy storage systems, including solid-state microbatteries and supercapacitors, is also critical for enhancing energy storage capacity without increasing the physical footprint [312], [313], [314], [315].

B. INTELLIGENT AND CONTEXT-AWARE COMMUNICATION

As WSNs are deployed in increasingly dynamic and heterogeneous environments, communication protocols must evolve to become intelligent and context-aware. The integration of ML at the protocol level will enable adaptive mechanisms that adjust transmission power, routing paths, and frequency bands based on network conditions and application demands [316], [317].

TABLE 9. Comparative analysis of challenges, solutions, and impact in WSNs.

Challenge	References	Key Issues	Proposed Solutions	Impact/Applications
Energy Limitations and Power Optimization	[267] [268] [269] [270] [271]	Limited energy availability, dependency on environmental conditions	Dynamic power optimization techniques, hybrid energy harvesting, stochastic energy availability models	Long-term WSN deployments in environmental monitoring, smart cities
Network Scalability and Congestion Control	[272] [273] [274] [275] [276]	Increased congestion and packet collisions in large-scale networks	Hierarchical clustering, adaptive clustering using ML models, cross-layer optimization	Large-scale industrial IoT, urban smart grids, and agriculture
Environmental Interference and Resilience	[277] [278] [279] [280] [281]	Communication degradation due to electromagnetic interference or signal attenuation in harsh environments	Redundancy and self-healing algorithms, autonomous UAV-assisted deployment strategies	Industrial monitoring, underwater WSNs, disaster management
Security Vulnerabilities	[282] [283] [284] [285] [286] [287]	Susceptibility to eavesdropping, jamming, sinkhole, and other attacks; impracticality of PoW blockchains for WSNs	Lightweight cryptographic algorithms (e.g., ChaCha20, SPECK), blockchain-based trust mechanisms, application-specific consensus protocols like PoS and reputation-based schemes	Critical infrastructure, healthcare, and defense; decentralized trust in resource-constrained WSNs
Application-aware Security and Privacy	[288] [289]	Domain-specific security and privacy concerns in healthcare, smart homes, and industrial WSNs	Data anonymization, selective disclosure, user authentication, and resource-aware protocols adapted to context-sensitive needs	GDPR/HIPAA-compliant healthcare, cyber-physical protection in industry, privacy-aware smart homes
Data Overload and Real-Time Analytics	[290] [291] [292] [293] [294]	Overwhelming data volumes from multi-modal sensing systems	Edge computing, FL models, efficient data preprocessing	Healthcare analytics, industrial automation, smart city applications
Standardization and Interoperability	[295] [296] [297] [298]	Lack of uniform protocols complicating multi-vendor system integration	Adoption of IPv6-based protocols (e.g., 6LoWPAN, CoAP), middleware solutions for unified device interfaces	Smart city and industrial IoT systems
Latency and QoS Constraints	[299] [300] [301] [302] [303]	High latency and inconsistent QoS in delay-sensitive applications	Predictive scheduling algorithms, TSN, energy-efficient communication protocols	Autonomous vehicles, industrial automation
Cost-Efficiency and Maintenance Challenges	[304] [305] [306] [307] [308] [309]	High deployment and maintenance costs, battery replacement issues	Predictive maintenance models, robotic systems for autonomous deployment and calibration	Remote environmental monitoring, space exploration, hazardous site deployments

For instance, AI-enhanced routing protocols can predict congestion and reconfigure network topology in real time, minimizing energy consumption and ensuring consistent data delivery. Protocols leveraging RL can optimize multi-hop communication in large-scale deployments, addressing challenges such as the energy hole problem and maintaining connectivity under high node density [318], [319], [320].

C. COLLABORATIVE AND FEDERATED INTELLIGENCE

The future of WSNs lies in collaborative intelligence, where sensor nodes work together to analyze and interpret data, moving beyond simple data aggregation. FL, a decentralized approach to ML, allows nodes to train models while retaining their local data collaboratively. This approach not only reduces the communication overhead but also ensures data privacy, making it suitable for sensitive applications like healthcare and defense [321], [322].

Research is needed to optimize FL algorithms for resource-constrained WSNs. Techniques such as sparsification, quantization, and model compression can reduce the computational and communication costs associated with collaborative intelligence. Furthermore, integrating edge servers to act as intermediaries between nodes and cloud systems will enhance scalability and processing capabilities [323], [324], [325].

D. QUANTUM SENSING AND COMMUNICATION

Quantum technologies hold significant potential for advancing the capabilities of WSNs. Quantum sensors, leveraging phenomena such as quantum entanglement and superposition, offer unprecedented sensitivity and accuracy in detecting physical parameters like magnetic fields, temperature, and gravitational waves. These sensors could transform applications in environmental monitoring, healthcare, and space exploration [326], [327], [328].

Quantum communication protocols, such as Quantum Key Distribution (QKD), promise unbreakable security by utilizing the principles of quantum mechanics to detect eavesdropping. Future WSNs could integrate quantum communication technologies to safeguard critical data transmissions, particularly in defense and infrastructure monitoring applications [329], [330].

While quantum technologies offer significant potential, their integration into WSNs remains a long-term goal. Current quantum communication systems are not yet miniaturized or energy-efficient enough to meet the stringent constraints of typical sensor nodes. Most advancements are still at the experimental stage or limited to high-end applications. Progress in integrated quantum circuits and photonics may eventually enable compact quantum modules suitable for specialized WSN deployments.

E. BIO-INSPIRED AND SELF-ADAPTIVE ARCHITECTURES

Bio-inspired architectures, modeled after biological systems, are emerging as a promising approach to designing resilient and efficient WSNs. Techniques such as swarm intelligence, genetic algorithms, and neural-inspired learning mechanisms allow networks to adapt dynamically to environmental changes and node failures [331], [332].

For example, swarm intelligence algorithms mimic the collective behavior of social insects to optimize routing paths and resource allocation in large-scale WSNs. Neural-inspired architectures enable nodes to process sensory inputs and adapt their operations based on local observations, improving fault tolerance and reducing energy consumption. These self-adaptive systems will play a crucial role in ensuring network longevity and robustness in unpredictable environments [333], [334].

F. WSNs IN EXTREME ENVIRONMENTS

Expanding the application of WSNs to extreme environments —such as deep oceans, polar regions, and outer space— presents unique challenges that require innovative solutions. Underwater WSNs, for instance, face high latency and limited bandwidth due to acoustic communication. Research into hybrid communication methods, combining acoustic, optical, and radio frequency technologies, aims to overcome these limitations [335], [336], [337].

In space exploration, WSNs must operate autonomously under extreme radiation and temperature variations. Radiation-hardened hardware and fault-tolerant algorithms will be critical for ensuring reliability. Additionally, lightweight protocols for interplanetary communication are necessary to enable real-time data exchange between planetary sensor networks and orbital spacecraft [338], [339], [340].

G. INTEGRATION WITH DIGITAL TWINS

Digital Twins, virtual replicas of physical systems, are poised to revolutionize WSN applications by enabling advanced simulations, predictive analytics, and real-time monitoring. WSNs will serve as the primary data acquisition layer for Digital Twins, continuously feeding them with real-world observations [341], [342].

Future research will focus on seamless integration between WSNs and Digital Twins, emphasizing low-latency communication and data consistency. For example, in smart city management, Digital Twins could simulate the impact of policy changes on traffic flow or energy consumption using real-time data from WSNs. Ensuring synchronization and scalability in these integrated systems will be key to their success [343], [344].

H. SECURITY AND PRIVACY IN EMERGING APPLICATIONS

As WSNs expand into applications involving sensitive data, such as personalized healthcare and smart infrastructure, ensuring security and privacy will become increasingly important. Future research will explore advanced cryptographic techniques, such as homomorphic encryption, which allows data to be processed without being decrypted, preserving privacy while enabling analytics [345], [346], [347].

Decentralized security frameworks like blockchain-based authentication and access control will also gain traction. These systems will eliminate single points of failure and enhance trust in multi-vendor deployments. Moreover, anomaly detection systems leveraging deep learning will identify security breaches in real time, reducing the response time to cyberattacks [348], [349], [350].

I. MULTI-MODAL AND MULTI-SENSOR FUSION

The integration of multi-modal sensing in WSNs enables richer datasets and more comprehensive analyses. Future trends will focus on efficient fusion techniques that combine data from heterogeneous sensors, such as image, acoustic, and environmental sensors, to improve decision-making accuracy [351], [352].

For instance, in precision agriculture, data from soil moisture sensors, drones, and weather stations can be combined using fusion algorithms to optimize irrigation and pest control. In healthcare, integrating physiological signals like ECG, SpO_2, and motion data will enhance diagnostic precision. Research into lightweight and real-time fusion algorithms will be crucial for extending these capabilities to resource-constrained WSNs [353], [354], [355].

Table 10 provides a comparative summary of future trends and research directions in WSNs, highlighting key features, research focus, and potential applications. The table categorizes advancements such as ultra-low-power and self-sustaining systems, intelligent and context-aware communication, collaborative and federated intelligence, quantum sensing, bio-inspired architectures, WSNs in extreme environments, integration with digital twins, and security enhancements.

Beyond listing emerging methodologies—such as RL-based energy schedulers, AI-driven communication protocols, FL for privacy-preserving analysis, and QKD for secure communication—the table also classifies the type of advancement required to realize each trend. These include developments in hardware design (e.g., energy-harvesting microsystems and quantum sensors), algorithmic innovation (e.g., context-aware and AI-driven protocols), architectural

TABLE 10. Future trends and research directions in WSNs.

Future Trend	References	Key Features	Research Focus/Proposed Solutions	Applications/Impact	Required Advances
Ultra-Low-Power and Self-Sustaining Systems	[310] [311] [312] [313] [314] [315]	Nanotechnology-enabled nanoscale energy harvesters, multi-source energy systems	RL-based energy schedulers, micro-energy storage systems (e.g., supercapacitors)	Long-term deployments in smart cities, remote monitoring, and energy-constrained environments	Hardware, Algorithm Design
Intelligent and Context-Aware Communication	[316] [317] [318] [319] [320]	AI-enhanced adaptive protocols for dynamic and heterogeneous environments	AI-based congestion prediction, RL for multi-hop communication optimization	Autonomous vehicles, industrial IoT, and large-scale network deployments	Algorithm Design, Communication Protocols
Collaborative and Federated Intelligence	[321] [322] [323] [324] [325]	Decentralized learning for privacy-preserving collaborative analysis	FL optimization via sparsification, model compression, and edge-cloud collaborative frameworks	Sensitive domains like healthcare, defense, and industrial predictive maintenance	Algorithm Design, System Architectures
Quantum Sensing and Communication	[326] [327] [328] [329] [330]	High-sensitivity quantum sensors; unbreakable quantum communication protocols	QKD, quantum-enhanced sensing for environmental monitoring	Environmental monitoring, defense, healthcare, and space exploration	Quantum Hardware, Theoretical Foundations
Bio-Inspired and Self-Adaptive Architectures	[331] [332] [333] [334]	Resilient, dynamically adaptive systems modeled after biological behaviors	Swarm intelligence for routing, neural-inspired learning for fault tolerance and energy efficiency	Resilient WSN deployments in unpredictable environments, e.g., disaster zones	Algorithm Design, Bio-Inspired Architectures
WSNs in Extreme Environments	[335] [336] [337] [338] [339] [340]	Hybrid communication methods for harsh environments	Radiation-hardened hardware, fault-tolerant algorithms, and hybrid acoustic-optical communication	Underwater networks, space exploration, and polar region monitoring	Environment-Resilient Hardware, Communication Protocols
Integration with Digital Twins	[341] [342] [343] [344]	Real-time simulation and analytics using WSN data	Low-latency communication, synchronization frameworks, and scalable data pipelines	Smart city management, industrial optimization, and predictive maintenance	System Architectures, Data Processing Algorithms
Security and Privacy in Emerging Applications	[345] [346] [347] [348] [349] [350]	Privacy-preserving data analytics for sensitive use cases	Homomorphic encryption, advanced cryptography, blockchain-based authentication	Personalized healthcare, critical infrastructure, and smart grids	Security Algorithms, Communication Protocols
Multi-Modal and Multi-Sensor Fusion	[351] [352] [353] [354] [355]	Integration of multiple sensing modalities such as image, acoustic, and environmental sensors	Development of efficient fusion techniques combining heterogeneous sensor data for improved decision-making accuracy	Precision agriculture, healthcare, industrial automation, and environmental monitoring	Data Fusion Algorithms

rethinking (e.g., bio-inspired and hybrid layered designs), and theoretical foundations (e.g., formal models for secure federated inference). This classification helps delineate whether progress depends on tangible engineering solutions or foundational conceptual breakthroughs, providing a clearer roadmap for researchers and practitioners. Collectively, these innovations are expected to enhance network efficiency, resilience, and scalability, benefiting applications such as smart cities, autonomous vehicles, industrial IoT, healthcare, defence, and environmental monitoring.

VII. CONCLUSION

WSNs have emerged as a cornerstone of modern technology, addressing critical challenges across diverse domains such as environmental monitoring, healthcare, industrial automation, and beyond. The advancements in energy harvesting, communication protocols, and data processing have enabled WSNs to transcend traditional limitations, offering scalable, reliable, and efficient solutions. However, the ever-expanding scope of WSN applications brings with it complex challenges and emerging research gaps that demand innovative approaches.

One of the central themes in the evolution of WSNs is achieving sustainability through energy efficiency and self-sustaining systems. Innovations such as multi-source energy harvesting and dynamic power management algorithms represent significant strides toward prolonging network lifespans, particularly in remote and resource-constrained environments. These developments ensure that WSNs can maintain functionality in diverse settings, from dense urban areas to extreme environments.

Integrating WSNs with emerging paradigms such as IoT, AI, and digital twins highlights their pivotal role in shaping future technologies. Adopting intelligent protocols and collaborative processing frameworks has transformed WSNs into data collection networks and platforms for real-time analytics and autonomous decision-making. FL and edge intelligence exemplify the potential for distributed processing, ensuring that WSNs remain viable even in

scenarios where centralized resources are unavailable or impractical.

Despite these advancements, critical challenges remain. Scalability issues in dense networks, dynamic topology management, and robust security are persistent obstacles that must be addressed through cutting-edge research. Security, in particular, remains a significant concern, as WSNs are inherently vulnerable to attacks due to their resource constraints. Developing lightweight cryptographic methods and blockchain-based trust mechanisms is essential for safeguarding sensitive data in applications such as healthcare, defense, and industrial automation. Similarly, quantum communication and bio-inspired architectures are set to redefine the boundaries of what WSNs can achieve regarding reliability and adaptability.

This study contributes a concise review of the current landscape of WSNs, structured around three dimensions: (i) domain-specific applications, (ii) technological innovations such as energy-aware and AI-integrated systems, and (iii) challenge-driven analysis across areas including energy efficiency, scalability, and security. Furthermore, comparative and case-based tabular summaries are presented to bridge theoretical advancements with real-world implementations.

Future research should prioritize the development of hybrid energy harvesting models with adaptive power scheduling, lightweight and context-sensitive security protocols, and resilient WSN frameworks capable of autonomous reconfiguration in extreme environments such as underwater, polar, and space missions. In parallel, optimized 5G-enabled architectures, fog-assisted intelligence, and seamless interoperability must be advanced to enhance real-time data processing and reduce latency. Cross-domain integration with Digital Twins, FL, and self-organizing systems will be critical to realizing the full potential of next-generation WSNs.

The trajectory of WSN development underscores their potential as transformative enablers across scientific, industrial, and societal domains. By addressing these open challenges and embracing interdisciplinary collaboration, WSNs are poised to become integral to the next wave of technological innovation across both Earth-bound and remote environments.

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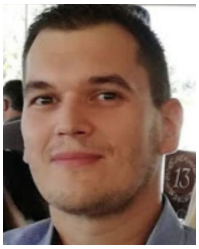
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